

Name: _____

GCSE Biology Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	7	11	19	27	36	45	54
2018	6	10	17	27	37	47	57
2019	12	16	23	30	37	44	51
2022	5	7	14	23	32	41	50
2023	10	14	19	28	38	48	58

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3400UA0-1



BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

HIGHER TIER

WEDNESDAY, 14 JUNE 2017 – MORNING

1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	12	
3.	5	
4.	6	
5.	8	
6.	12	
7.	7	
8.	10	
9.	6	
10.	6	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

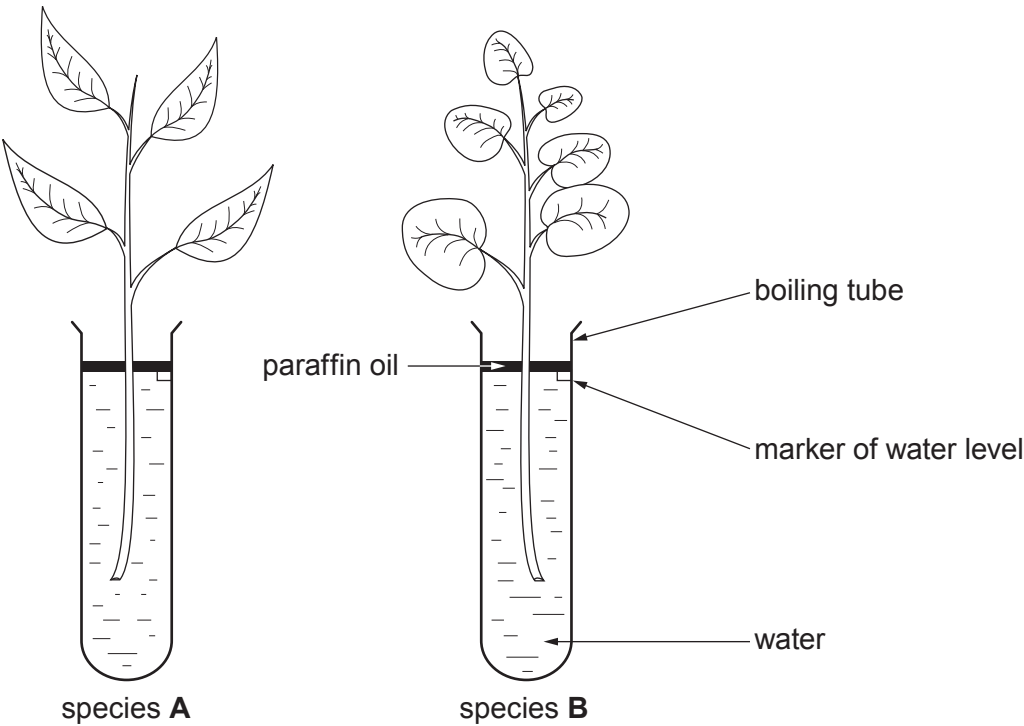
- Use black ink or black ball-point pen.
- Write your name, centre number and candidate number in the spaces at the top of this page.
- Answer **all** questions.
- Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

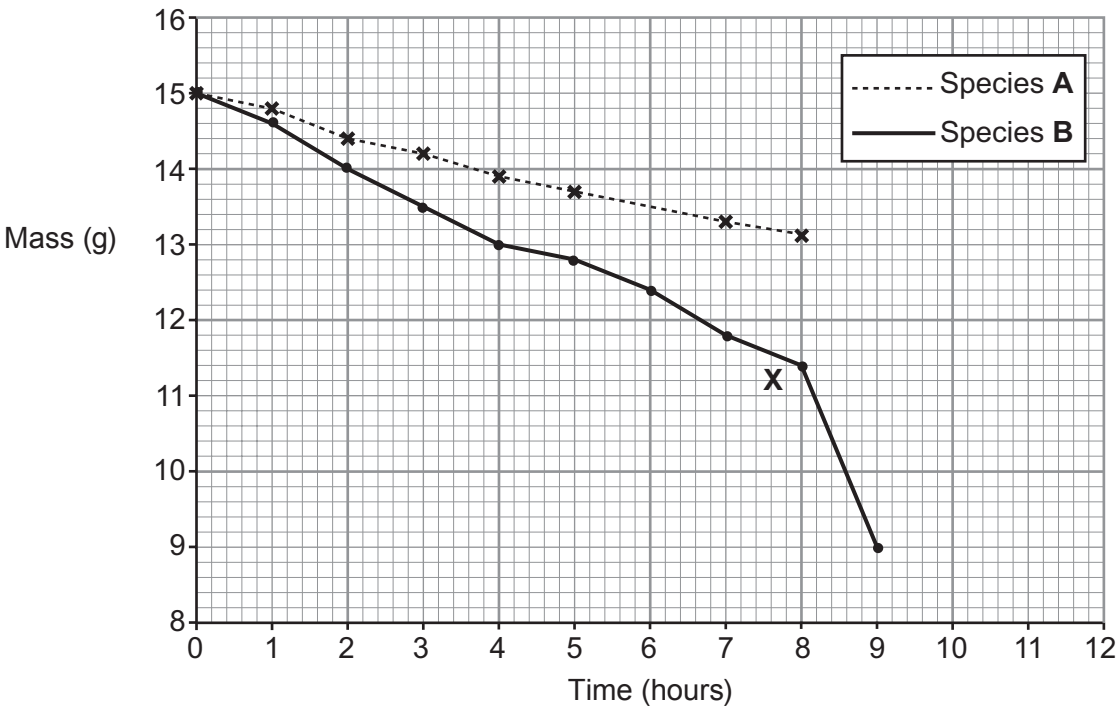
The number of marks is given in brackets at the end of each question or part-question.
Question **9** is a quality of extended response (QER) question where your writing skills will be assessed.

Answer all questions.

1. Shoots from two different species of plants, **A** and **B**, were placed in water in boiling tubes as shown below.



Both species were kept in the same conditions and their mass recorded at hourly intervals for eight hours. The results were recorded as line graphs shown below.



- (a) (i) Use the data to help you calculate the difference in the loss in mass between species **A** and **B** at 8 hours. [1]

Difference in loss of mass = g

- (ii) State the term given to the process responsible for this loss of mass in plants. [1]

- (iii) Suggest **two** reasons for the difference in loss of mass between the two different species, **A** and **B**. [2]

- (iv) State the purpose of the paraffin oil in the boiling tubes. [1]

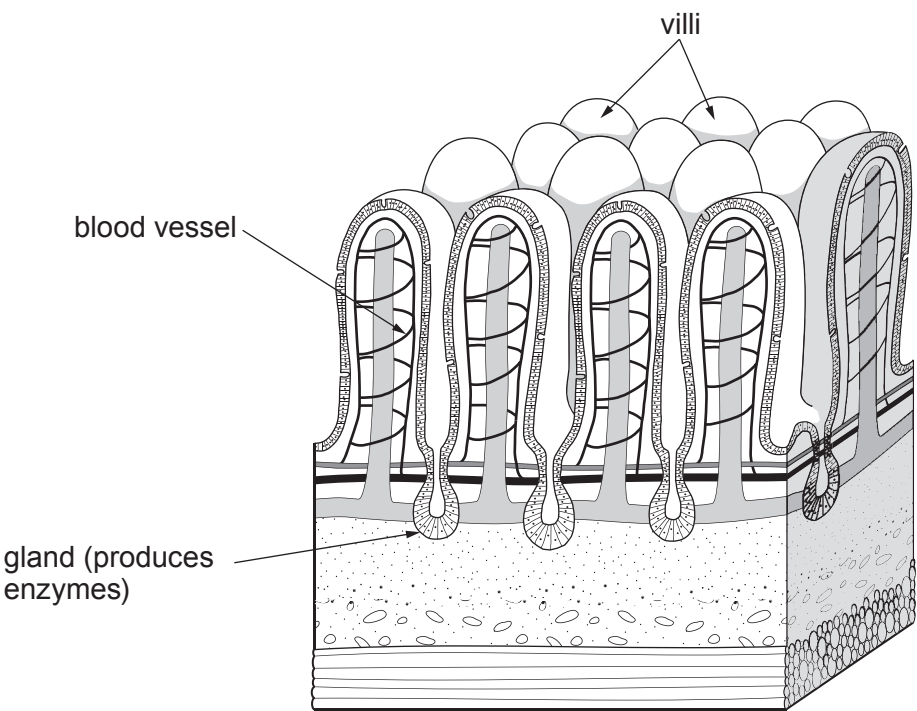
- (b) Which **one** of the following treatments was applied to species **B**, at point **X** on the graph, to produce the result shown at 9 hours?

Underline the correct answer. [1]

- Enclosed in a bell jar
- Upper surface of leaves coated with Vaseline
- Half the leaves removed
- Air directed at the leaves by a fan

- (c) What would you expect to happen to the rate of loss of mass if species **A** and **B** had roots? Give a reason for your answer. [2]

2. The drawing shows the lining of part of the digestive system.



(a) Name the part of the digestive system that has this lining. [1]

(b) If the surface area of a single villus is 5 mm^2 , calculate the total surface area of all the villi shown. Assume that all the villi are complete and include a unit in your answer. [1]

Total surface area =
unit =

(c) State **two** functions carried out in this part of the digestive system. [2]

Function 1.
Function 2.

(d) State **one** feature of the digestive system which helps this part carry out each function named in (c). [2]

Function 1.

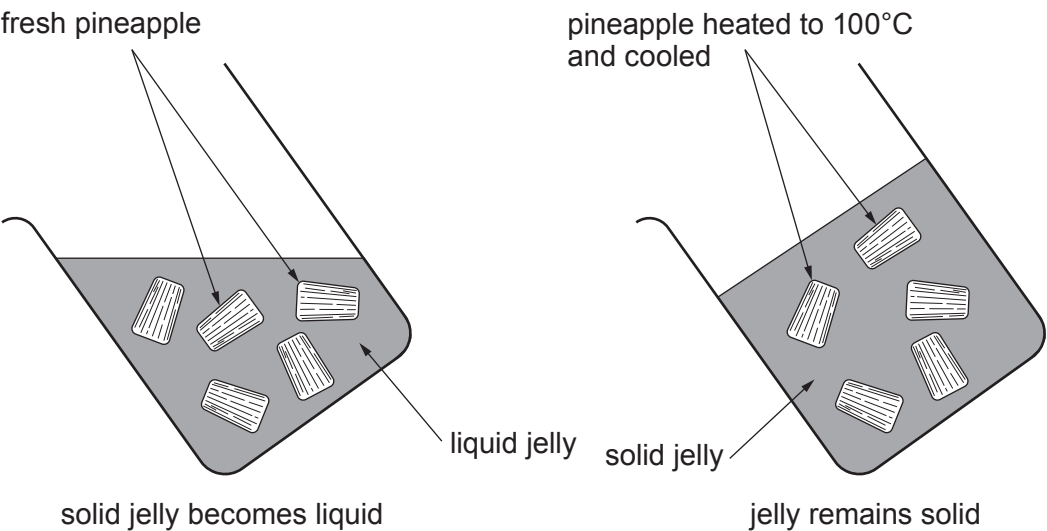
Function 2.

(e) Fresh pineapple is sometimes added to meat to make it tender. Fresh pineapple contains a protease enzyme.

Alan investigated how this took place. He used the following:

- fresh pineapple
- pineapple that had been heated to 100 °C and cooled
- a solid jelly made from protein called gelatin

Two samples of solid jelly were made and the pineapple was pushed into them. After 24 hours the results were shown below:



Explain the results with:

(i) fresh pineapple; [3]

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(ii) pineapple heated to 100°C and cooled. [3]

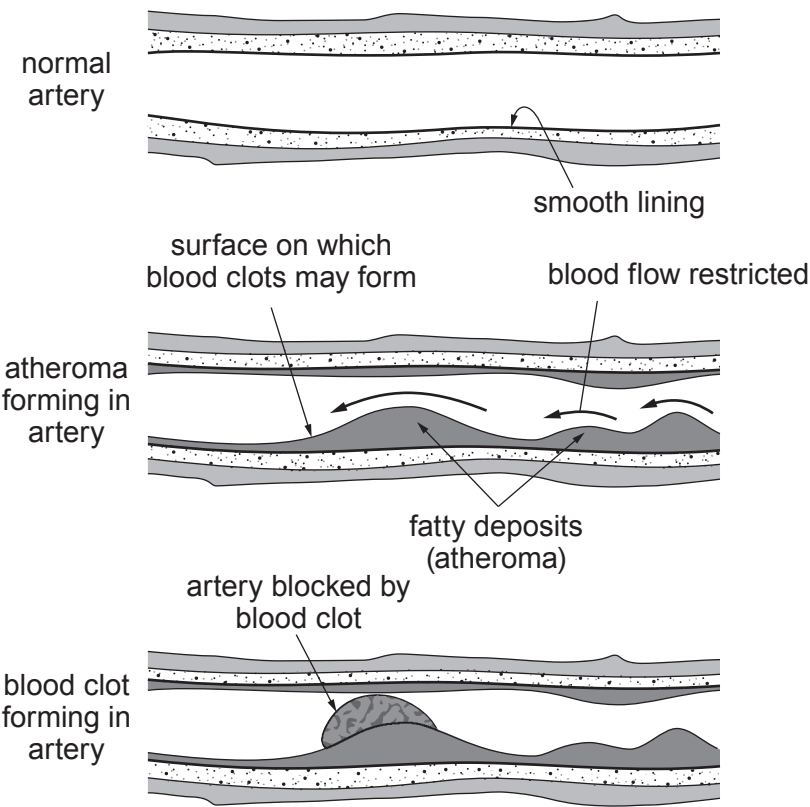
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3. The diagrams show fatty deposits (an atheroma) building up in an artery. One cause of this is high cholesterol. This may eventually cause a blood clot to form.



(a) Explain how an unbalanced diet and lack of exercise could lead to an atheroma formation. [2]

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(b) Name a group of drugs which reduces the chance of build-up of fatty deposits in arteries. [1]

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(c) Explain how an atheroma may lead to a heart attack. [2]

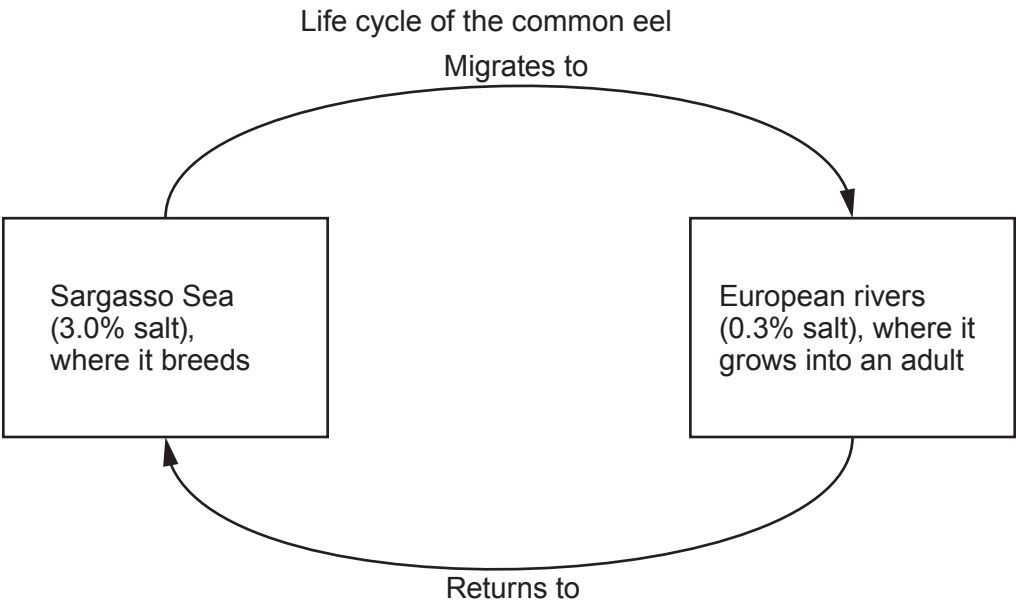
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Examiner
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4. The common eel (*Anguilla anguilla*), spends some of its life in fresh water and some of its life in sea water. It breeds in the Sargasso Sea near Central America and migrates to European rivers where it becomes an adult. After several years, it becomes sexually mature and returns to the Sargasso Sea to breed.



- (a) Explain why osmosis could be a problem to the eels when they return from fresh water (0.3% salt) to sea water (3.0% salt). [3]

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(b) Eels are able to absorb salt against a concentration gradient.

Name the process responsible for this and the **two** chemicals that are needed for the process. [3]

Name of process

Chemicals 1

2

6

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5. (a) The table shows a comparison for two athletes, Tomos and Jeremy, who ran in races of different distances.

Athlete	Distance of race (m)	Oxygen needed in the race (dm ³)	Oxygen entering blood in the race (dm ³)
Tomos	100	10	0.5
Jeremy	10 000	150	134.0

- (i) The difference between the oxygen needed and the oxygen actually entering the blood during the race is the oxygen debt. Calculate the oxygen debt for [1]

Tomos dm³

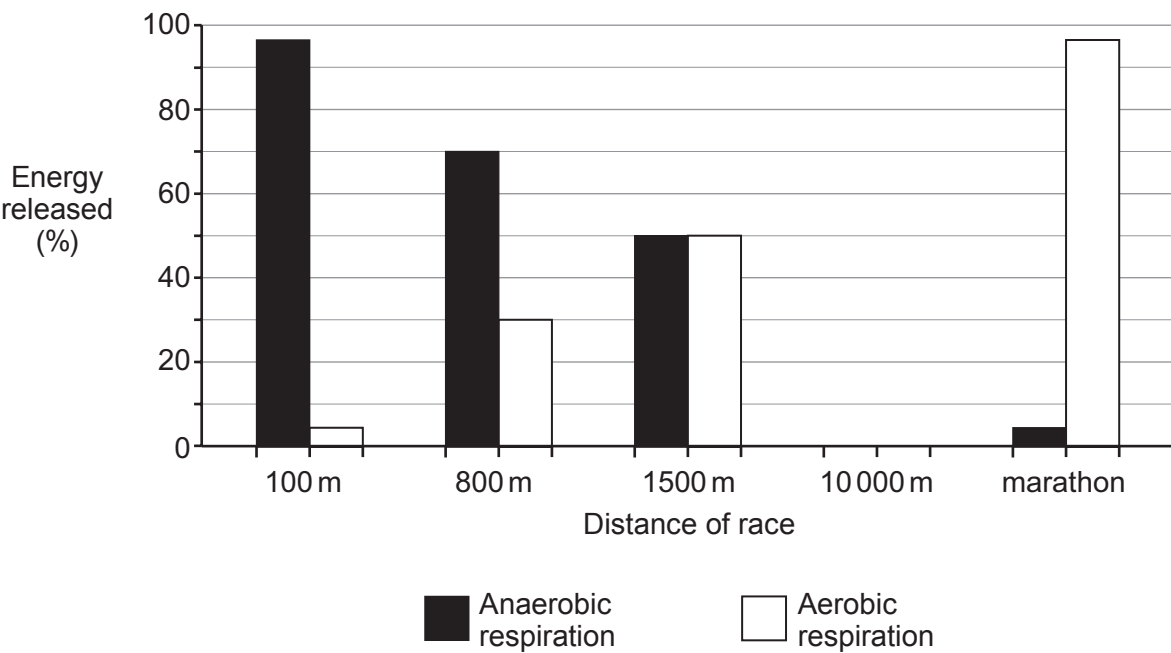
Jeremy dm³

- (ii) When the race was over, both athletes continued to breathe more rapidly and more deeply than normal for some time. Give the reason for this. [1]

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- (b) The bar chart shows the percentage of energy released by aerobic and anaerobic respiration for athletes running different distances.



Examiner
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- (i) What can be concluded from the data shown in the table and the bar chart for the athlete running 100 m? [1]

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- (ii) **On the bar chart opposite, draw bars** that you would expect for an athlete running in a 10 000 m race. [2]

- (c) Write the word equations to represent:

- (i) Aerobic respiration in human muscle tissue; [1]

.....

(ii) Anaerobic respiration in human muscle tissue. [1]

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- (d) Why is aerobic respiration more efficient than anaerobic respiration? [1]

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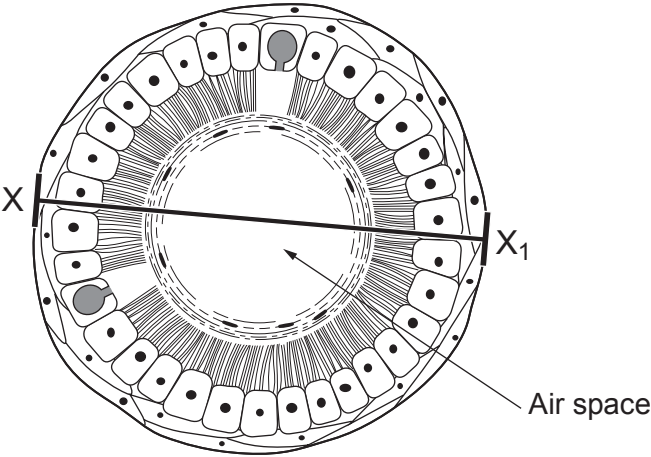
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8

6. The diagram below shows a cross section of a healthy bronchiole.

Diagram A

A healthy bronchiole

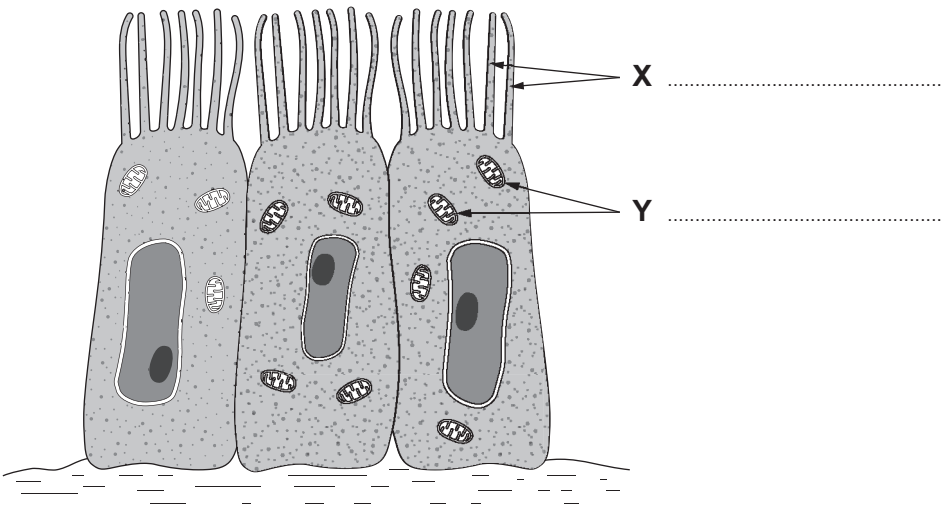


- (a) If the magnification of diagram A is x50, use the line X–X₁ to calculate the actual diameter of the bronchiole. [1]

Diameter = mm

- (b) (i) Diagram **B** below shows a highly magnified close-up of three cells of the lining of the bronchiole.
Label parts X and Y on diagram B. [2]

Diagram B



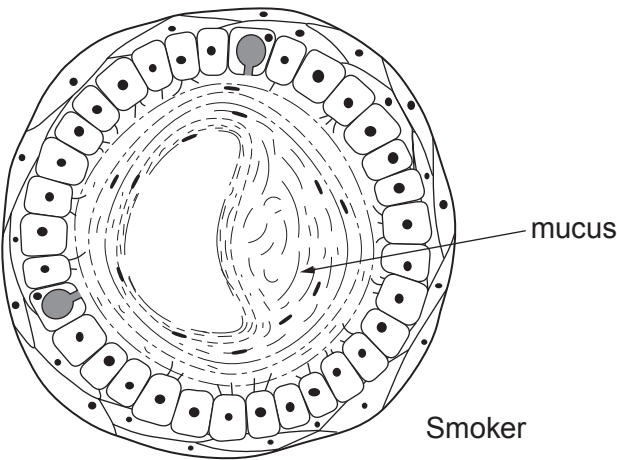
- (ii) Explain how the process taking place in part Y allows part X to carry out its function. [3]

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The following diagram shows a section through a bronchiole from a person who had smoked for 20 years.



- (c) What would you expect to find in the mucus of this person? [1]

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- (d) Nicotine is the addictive substance in tobacco. People can be helped to stop the habit of smoking by giving them nicotine replacement therapy (NRT). The table gives the results of the trials of different types of NRT. The placebo is a neutral substance which does not contain nicotine.

	Smokers given NRT		Smokers given a placebo	
Type of NRT	Numbers of smokers in trial	Percentage of smokers who gave up smoking	Numbers of smokers in trial	Percentage of smokers who gave up smoking
chewing gum	7708	20	9319	12
patch on skin	7387	14	5969	8
nasal spray	448	24	439	12
inhaler	490	14	486	8
tablet	243	20	245	13

- (i) Why was the placebo used in these trials? [1]

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- (ii) Use the results to state the type of NRT in which scientists would have the greatest confidence. Give a reason for your answer. [2]

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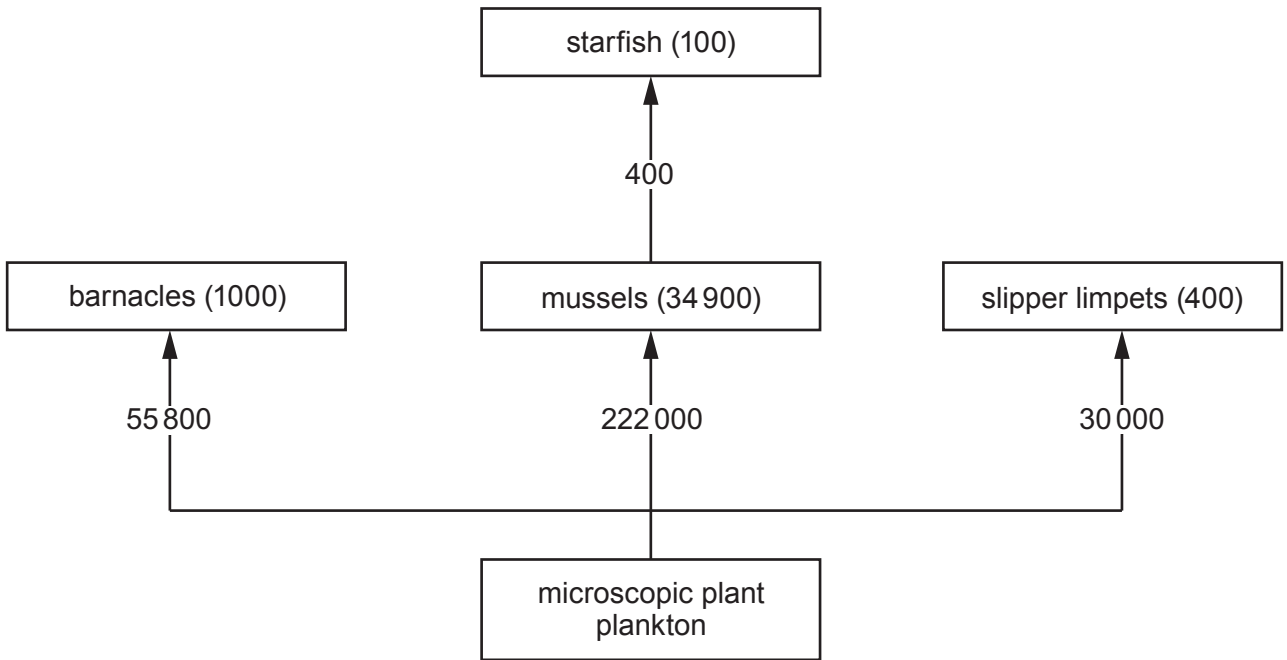
- (iii) Which type of NRT was most effective? Give a reason for your answer. [2]

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7. The flow diagram shows the transfer of energy between organisms on a rocky shore. Numbers on the arrows show the energy available to the organisms in kJ per m^2 per year. Numbers in brackets show the energy that becomes part of the biomass of the organisms in kJ per m^2 per year.



The energy efficiency of an organism is a measure of how much of the energy available to the organism becomes part of its biomass. It is calculated by using the following equation:

$$\% \text{ energy efficiency} = \frac{\text{energy that becomes part of biomass}}{\text{energy available}} \times 100$$

- (a) (i) Calculate the energy efficiency of the starfish. [2]

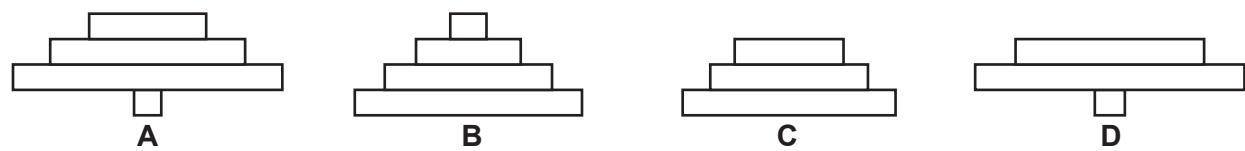
energy efficiency = %

- (ii) Calculate the percentage of the energy entering the mussels that enters the starfish to **three** significant figures. Show your working. [3]

percentage of the energy entering the mussels = %

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(b) The diagrams show pyramids of numbers for four ecosystems.



Which **one** of the pyramids of numbers represents the flow diagram of this rocky sea shore? State a reason for your answer. [2]

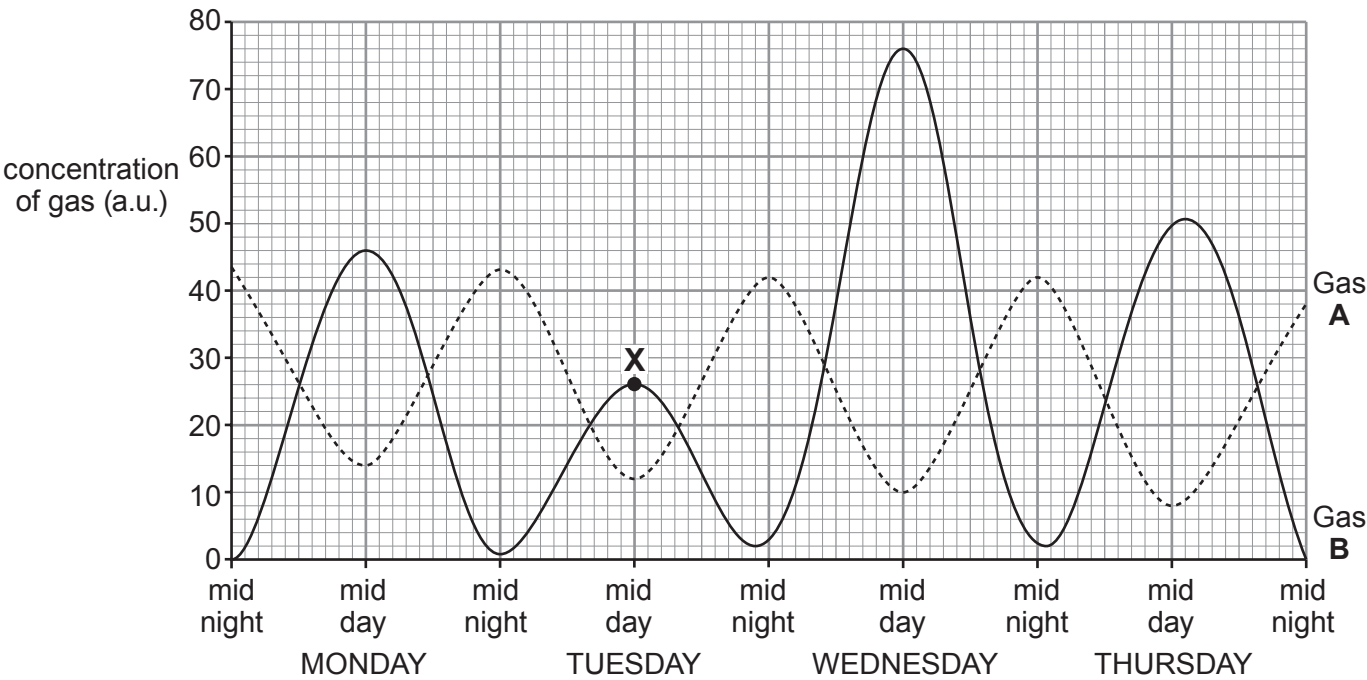
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8. The graphs below show changes in the concentration of two gases, **A** and **B**, in a rock pool in St Brides Bay, Pembrokeshire over a 4-day period. The rock pool has a high biomass of plants.



(a) (i) Identify gases **A** and **B**. [2]

Gas **A**

Gas **B**

(ii) Explain the results at midday for gas: [4]

A;

.....

B.

.....

(iii) Suggest a reason for the lowest peak at point **X**. [1]

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(iv) Calculate the mean maximum concentration of gas **B** over 4 days.

[2]

Examiner
only

mean maximum concentration of gas **B** = a.u.

(b) Name the limiting factor that can be deduced from the data in the graph.

[1]

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Examiner
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9. Urea is an animal waste substance which contains nitrogen. Explain how nitrogen in the urea from cattle in a field of grass is recycled to become protein in beef. [6 QER]

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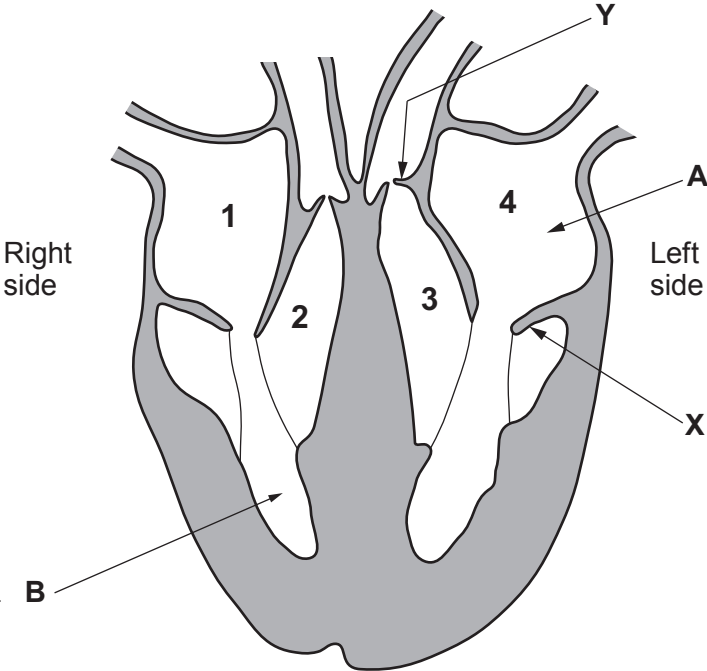
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10. The diagram shows a section through the heart.



- (a) (i) Label parts **A** and **B** on the diagram. [2]
- (ii) Draw arrows at points **1**, **2**, **3** and **4** on the diagram to show the direction of flow of blood. [2]
- (b) The table shows the pressure readings in parts of the heart while it is pumping blood.

	Pressure when left ventricle is contracting (kPa)	Pressure when left ventricle is relaxing (kPa)
left ventricle	16.0	0.7
left atrium	0.4	1.1
aorta	12.0	10.7

- (i) Use the data in the table to state the pressure in the following parts when the valve labelled **X** is open. [1]

Part	Pressure (kPa)
left ventricle	
left atrium	

- (ii) Use the data in the table to state the pressure in the following parts when the valve labelled **Y** is open. [1]

Part	Pressure (kPa)
left ventricle	
left atrium	

END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3400UA0-1



BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

HIGHER TIER

MONDAY, 11 JUNE 2018 – MORNING

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	10	
3.	7	
4.	13	
5.	10	
6.	9	
7.	10	
8.	11	
Total	80	

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **8(c)** is a quality of extended response (QER) question where your writing skills will be assessed.



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Answer all questions.

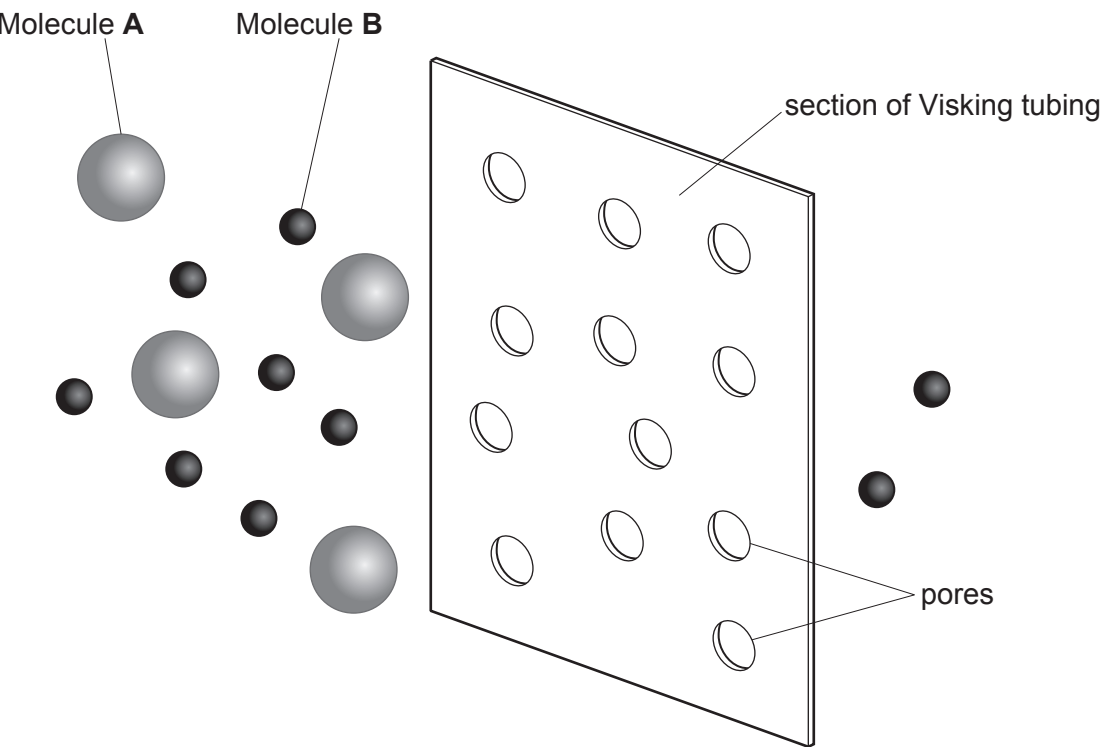
1. (a) What is meant by the term selectively permeable membrane? [1]

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Visking tubing can be used as a model of the cell membrane.

The diagram below shows a section of Visking tubing working in the same way as a selectively permeable membrane.



Use the diagram above and your own knowledge to answer the following:

(b) (i) State the process by which molecules could pass through the Visking tubing. [1]

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(ii) Identify which molecules pass through the Visking tubing. Explain your answer. [2]

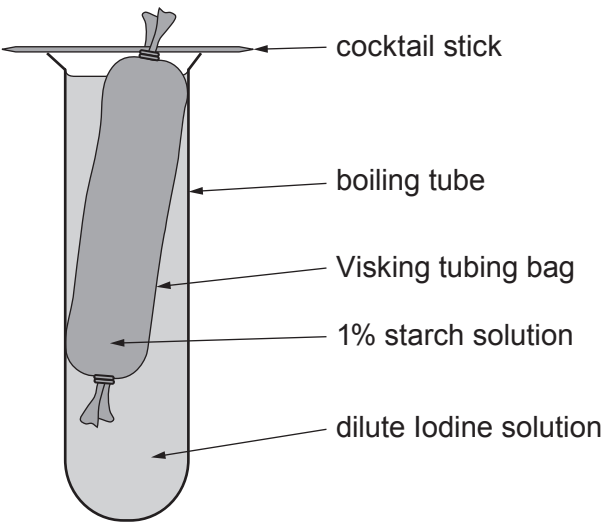
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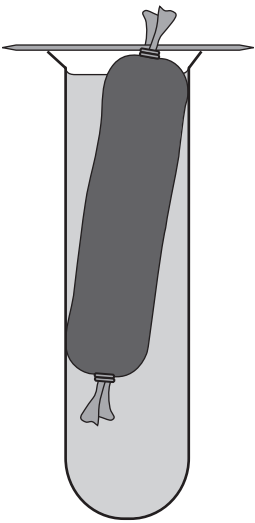
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(c) Students were instructed to set up the following apparatus.



The appearance of the apparatus after 15 minutes is shown below.



(i) Explain why the colour of the contents inside the Visking tubing turned blue black. [3]

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Examiner
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(ii) Explain why the colour of the Iodine solution in the boiling tube did not change. [2]

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(d) Name **one** substance required for respiration that would pass into a cell. [1]

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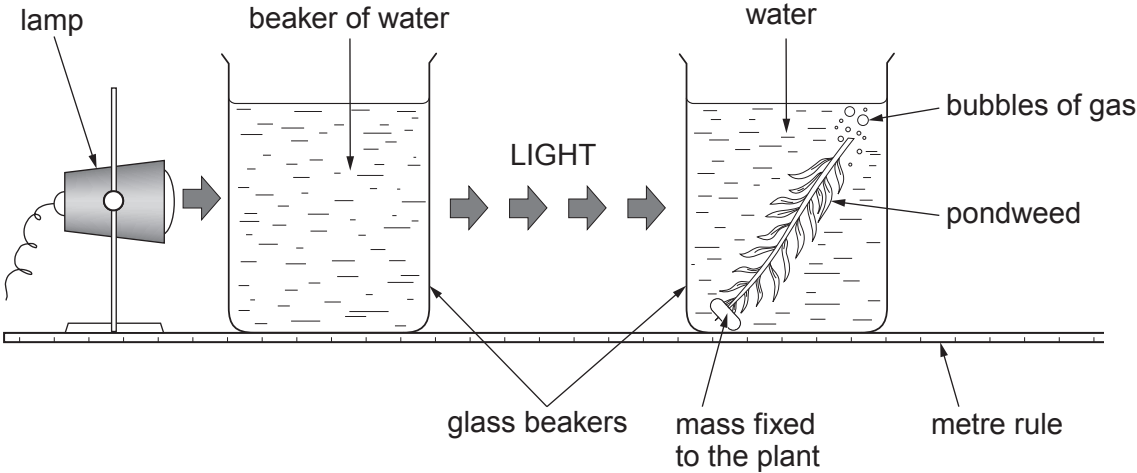
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2. (a) (i) Write the **word** equation for photosynthesis. [2]

(ii) State the name of the pigment present in plant cells which absorbs light. [1]

Phoebe and Adam used the apparatus below to study the rate of photosynthesis in the pondweed (*Elodea sp.*).



The number of gas bubbles per minute produced by the pondweed was counted at different distances from the light.

The experiment was carried out three times at each distance.

The results are shown below. Means were calculated to the nearest whole number.

Distance of lamp from pondweed (cm)	Number of bubbles per minute			
	Test 1	Test 2	Test 3	Mean
10	19	32	25	25
20	14	20	20	18
30	11	15	17	
40	7	10	13	10
50	5	9	11	8



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(b) **Complete the table** opposite by calculating the mean number of bubbles for a distance of 30 cm. **Write your answer in the table.** [2]

(c) State the relationship between the distance of the lamp from the pondweed and number of bubbles produced per minute. Explain your answer. [3]

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(d) Explain why a beaker of water was placed between the lamp and the pondweed. [1]

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(e) State how you could improve the accuracy of this investigation. [1]

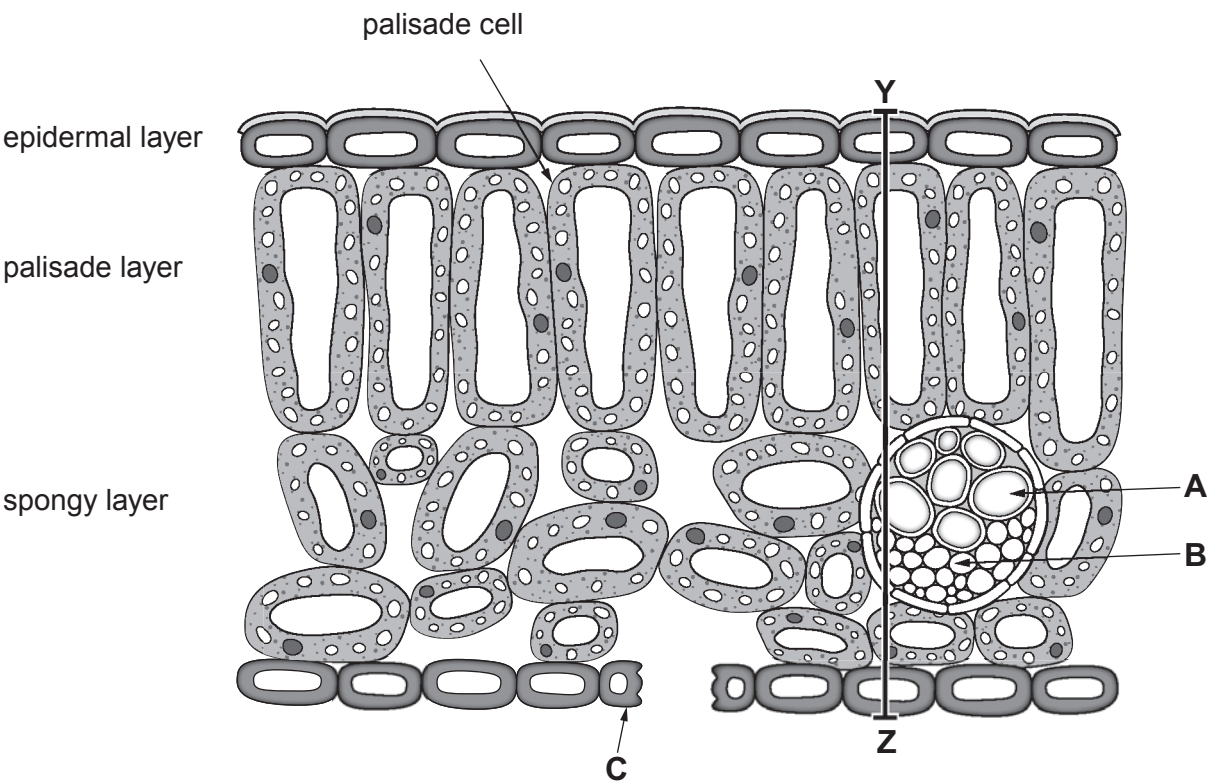
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3. The diagram below shows a section through a leaf of a sunflower (*Helianthus sp.*).



(a) State the name of structures **A**, **B** and **C**. [2]

A

B

C

(b) (i) The palisade cell is an example of a specialised cell.
What is the advantage to the plant of having specialised cells? [1]

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(ii) The leaf is considered to be an organ of the plant. What is meant by this statement? [1]

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(c) (i) Measure the length of line **Y – Z** on the drawing.

[1]

Length = mm

(ii) The actual length of **Y – Z** on the diagram opposite is 225 μm .
Calculate the magnification of the drawing.
(1 mm = 1000 μm)

[2]

= $\times$

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09

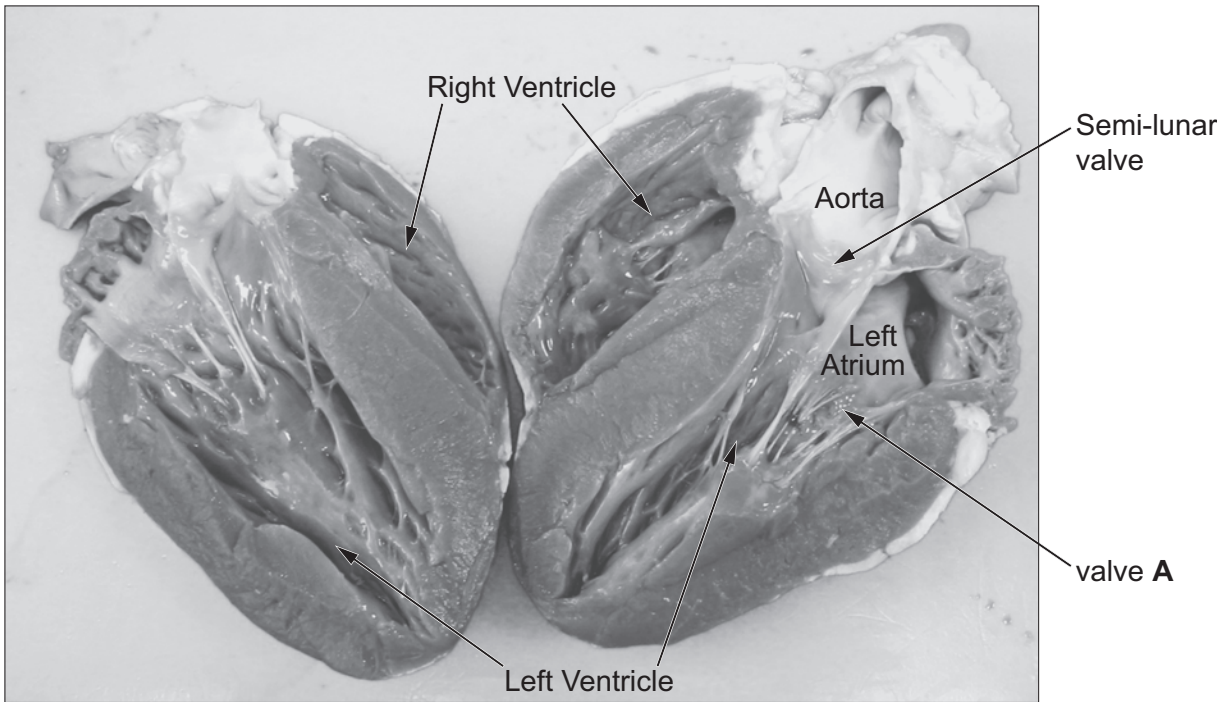


4. (a) State what is meant by a double circulatory system. [2]

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The photograph below shows a section through the heart.



(b) Use the picture above and your own knowledge to identify valve A. [1]

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(c) Explain how the semi-lunar valve, shown in the photograph, helps to ensure a one-way flow of blood through the heart. [3]

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Examiner
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(d) The walls of the left ventricle are thicker than the walls of the right ventricle. Explain the significance of this difference. [2]

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In 2016, a research group at Imperial College London reported that weekly exercise seemed to increase the thickness of the walls of the ventricles. This effect could be mistaken for serious heart disease even though the individuals are healthy.

(e) Suggest an explanation for the effect of exercise on the thickness of the walls of the ventricles. [2]

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Continued on next page

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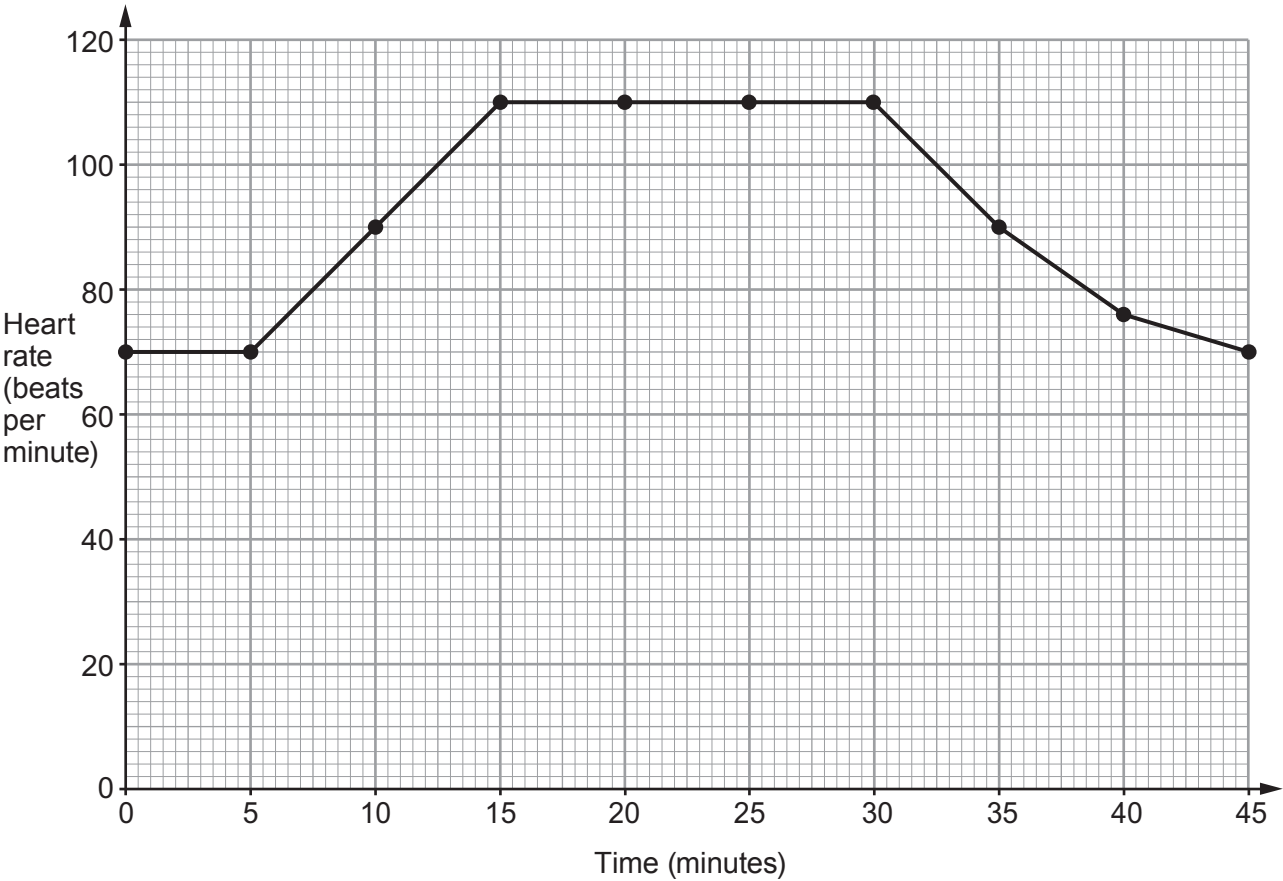


(f) Cardiac output is a measurement of the volume of blood pumped by a ventricle in one minute. It can be calculated as follows:

Cardiac output = volume of blood in ventricle $\times$ heart rate

The volume of blood in a ventricle of an average adult human is 70 cm^3 .

The graph below shows the heart rate of an individual before, during and after a session on an exercise bike.



(i) Calculate the cardiac output at 5 minutes and 20 minutes. [1]

5 minutes = cm^3/min

20 minutes = cm^3/min



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(ii) Calculate the percentage increase in cardiac output between 5 and 20 minutes. [2]

percentage increase in cardiac output = %

13

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5. The following article appeared on the BBC news website on the 13th of April 2010.

A “perfect storm” of nutrients from sewage effluent and unusual weather conditions has been blamed for a 2009 outbreak of toxic algae in a lake.

- 1 Llyn Padarn at Llanberis was closed to leisure users for much of last summer.

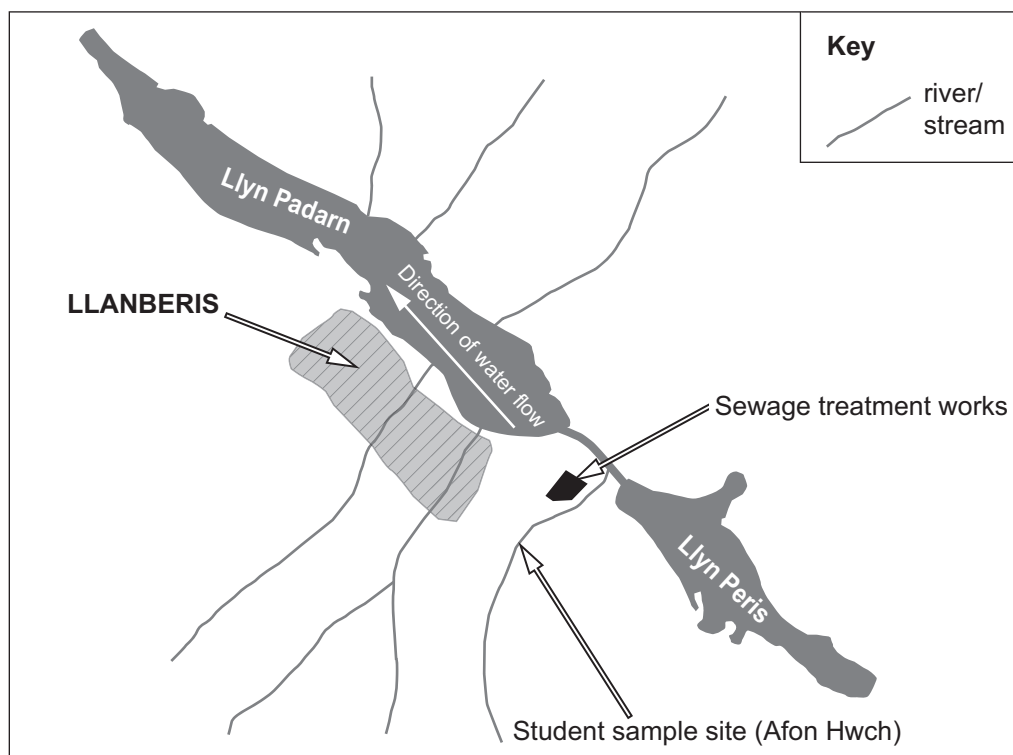
A report commissioned by Environment Agency Wales calls for stricter limits on effluent discharged from the local sewage treatment works.

- 5 The report by Professor Glen George described the warm spring, followed by a still June and a wet July and August as “the perfect storm”.

He said the weather conditions, combined with nutrients in the lake, not only created the right conditions for the algal bloom to form but also meant it continued through the summer.

- 10 His report makes a number of recommendations for further scientific studies of the lake and more work to find out how the management of the sewage treatment works in Llanberis is affected by heavy rain.

The map below shows Llyn Padarn and the surrounding area.



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Llyn Padarn is also home to a rare breed of fish called the Arctic char fish (*Salvelinus alpinus*). The char feeds on insects found on the water’s surface, fish eggs and snails found on the lake bottom.

(a) Explain how the **unusual weather** conditions in 2009 referred to in lines 4 and 5 contributed to the outbreak of algae. [2]

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(b) Explain why the local fishermen thought that the algal blooms could lead to a decrease in the population of the arctic char. [3]

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Students from a local school carried out a survey on the Afon Hwch, a river that flows into Llyn Padarn, during July of 2009. The site of the survey is shown on the map on page 14. The results are recorded below:

Number of organisms						
stonefly nymph	mayfly nymph	caddis fly larvae	blood worm	leeches	sludge worms	rat tailed maggot
3	17	7	0	5	0	0

The following table can be used to assess the condition of the water.

Organism present	Levels of pollution	Oxygen concentration
no visible life	very high	no oxygen
rat tailed maggot	high	low
sludge worm		
leeches	moderate	moderate
blood worm		
caddis fly larvae	low	moderate to high
mayfly nymph	none	high
stonefly nymph		

(c) (i) State the scientific term which is used to describe organisms that are used to detect levels of pollution. [1]

(ii) Consider the information provided to describe the water conditions in the Afon Hwch. [2]



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(iii) It was recommended that further scientific studies were needed to show that the algal blooms were not caused by fertilisers used on the land around the lake. Describe how students could extend their survey to test this idea. [2]

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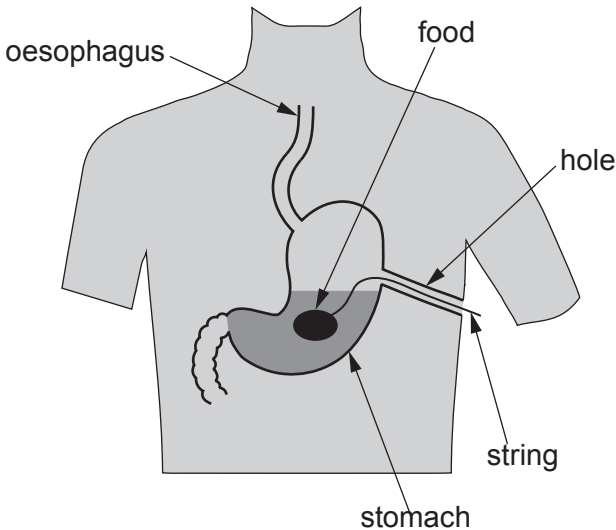
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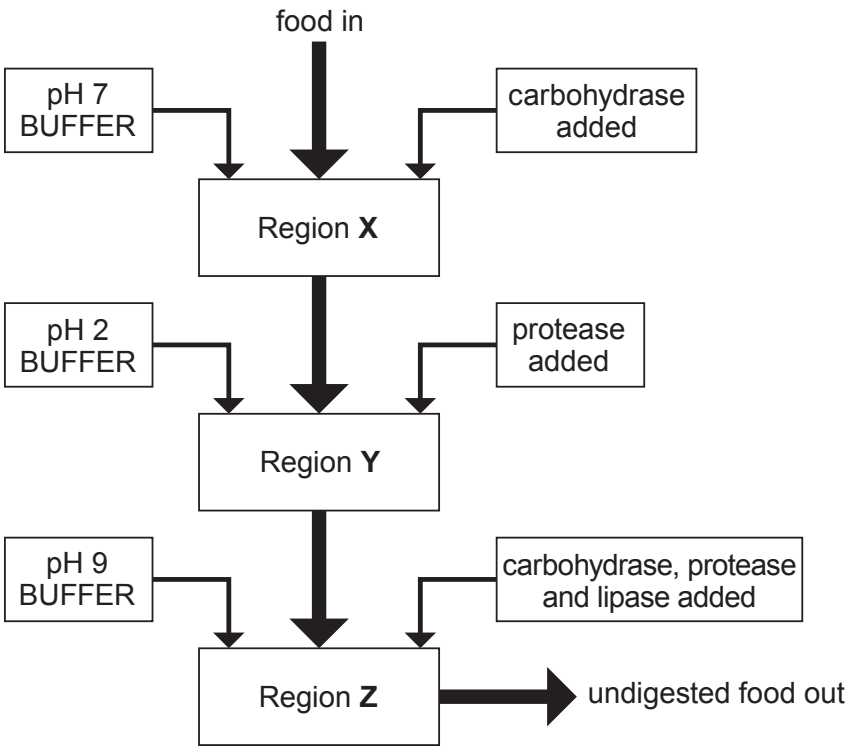
6. On June 6, 1822 Alexis St. Martin, was accidentally shot in the stomach from close range. Dr. William Beaumont treated his wound, but expected Alexis to die from his injuries. Alexis survived, but with a hole through his abdomen wall into his stomach that never fully healed.
- Dr. Beaumont began to carry out experiments on digestion by tying a piece of food to a string and inserting it through the hole into Alexis' stomach. Every few hours, Beaumont would remove the food and assess how much digestion had happened. Beaumont also extracted a sample of gastric juice from the stomach. Analysis showed that the gastric juice was acidic.

Fig 1 – Diagram showing Dr. Beaumont's experiment



Scientists can now follow the digestion of food in detail by using an artificial gut. The diagram below shows how an artificial gut works. (Note: a pH buffer is a chemical that keeps pH constant.)

Fig 2 – Flow chart of an artificial gut



Examiner
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(a) (i) State the region of the artificial gut that represents the stomach. [1]

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(ii) Explain why the pH of each region needs to be different. [2]

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(iii) State **one other** factor that would need to be controlled to ensure valid results from an artificial gut. [1]

.....

(iv) State the role of lipase in region Z. [1]

.....

(b) One of Dr. Beaumont's experiments involved inserting meat on a piece of string through the hole in Alexis' stomach and observing the time taken for the meat to disappear.

Explain why the meat disappeared. [2]

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(c) Suggest **two** reasons why scientists prefer to use an artificial gut rather than using human trials. [2]

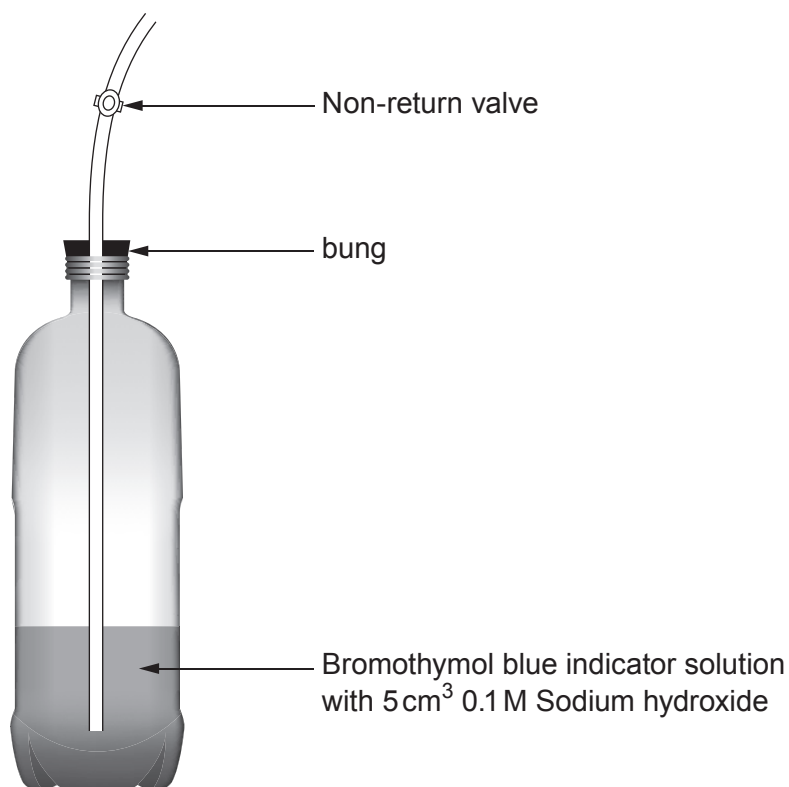
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7. A respirometer is a device used to measure the rate of respiration of a living organism by measuring its rate of exchange of oxygen and/or carbon dioxide. A simple respirometer used in a school laboratory is shown below. It can be used to estimate the volume of carbon dioxide in exhaled air.



Fact file

- Bromothymol blue indicator is green when neutral and blue when alkali.
- 5.6 cm³ of carbon dioxide will neutralise the sodium hydroxide in the respirometer.

The following equation can be used to calculate the volume of carbon dioxide produced per minute.

$$\begin{array}{ccccc} \text{Volume of carbon dioxide per minute} & = & \text{breathing rate} \times & \text{volume of carbon dioxide in one breath} \\ (\text{cm}^3/\text{minute}) & & (\text{breaths/minute}) & & (\text{cm}^3/\text{breath}) \end{array}$$



Use the information from the fact file opposite to answer the following questions.

- (a) (i) Alun wanted to investigate the effect of exercise on the volume of carbon dioxide breathed out. While resting, he takes 5 breaths over a period of 25 seconds to change the colour of the Bromothymol indicator from blue to green. Use the equation given in the fact file to calculate the volume of carbon dioxide produced per minute. [3]

volume of carbon dioxide per minute = cm³/minute

- (ii) Suggest a possible source of error in the experimental method that could lead to inaccurate results. [1]

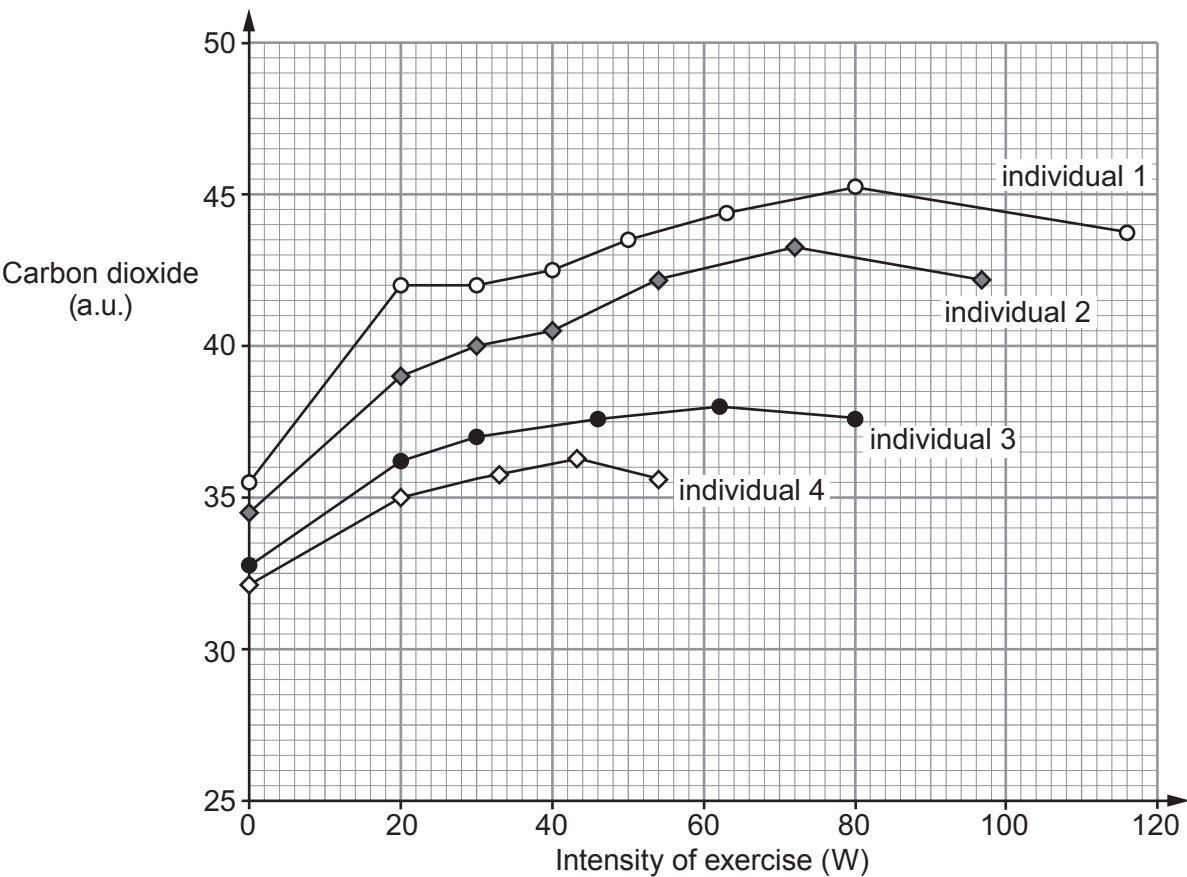
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- (b) A group of scientists monitored the proportion of carbon dioxide expired in 4 individuals. The individuals were tested on an exercise bike. They were allowed to rest for 4 minutes, and then exercised at an intensity of 20 watts for 4 minutes. After this time the intensity was increased 1 watt every 6 seconds. Individuals stopped exercising when they suffered a cramp in their leg muscles. A gas analyser was used to get an accurate measurement of the expired carbon dioxide level per breath.

The results of the investigation are shown below.



- (i) Explain the increase in carbon dioxide levels between an intensity of exercise 0 W and 20 W for all the individuals. [2]

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Examiner
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(ii) The decrease in carbon dioxide levels for all individuals at a higher intensity of exercise was linked to cramp in the muscles. Suggest an explanation for this. [2]

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(iii) Suggest which individual is the least fit. Give a reason for your answer. [1]

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(c) State why sports scientists working with athletes would measure oxygen consumption as well as carbon dioxide production. [1]

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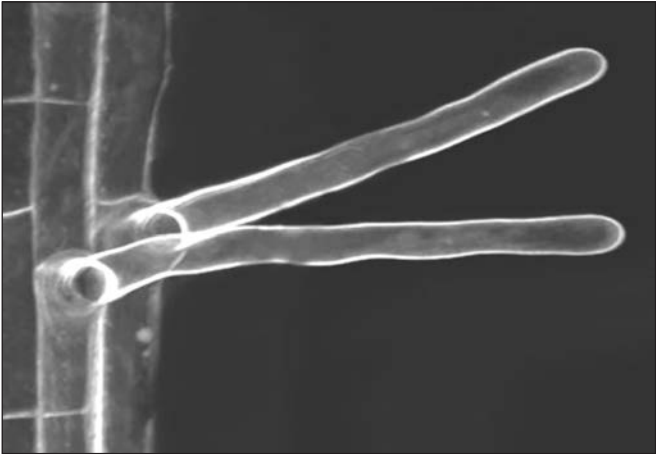
10



8. The pictures below show a young bean seedling (*Vicia faba*) root and an electron micrograph of root hair cells.



Root with root hairs
(left) and magnified
view of root hairs (right)



(a) State how the root hair cell is adapted for its function. [1]

.....

(b) Describe how water moves from the soil into a root hair cell. [4]

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Examiner
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The following is an extract from a UK government report from 2016:

“During the period from mid-December 2013 through until early February 2014 the UK suffered a spell of extreme weather, with a series of very large winter storms battering the country. Much of the UK suffered from extensive flooding during January 2014; this period saw the highest rainfall on record. The floods impacted on the agricultural sector through damage to or loss of established crops.”

Soil becomes waterlogged when water builds up and is unable to drain away. This reduces the oxygen levels available to root hair cells.

- (c) Explain why reduced oxygen levels for the root hair cells results in poor growth and yellowing of the leaves. [6 QER]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3400UA0-1



BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

HIGHER TIER

FRIDAY, 7 JUNE 2019 – AFTERNOON

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **7** is a quality of extended response (QER) question where your writing skills will be assessed.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	9	
3.	6	
4.	6	
5.	11	
6.	14	
7.	6	
8.	10	
9.	7	
Total	80	



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Answer all questions.

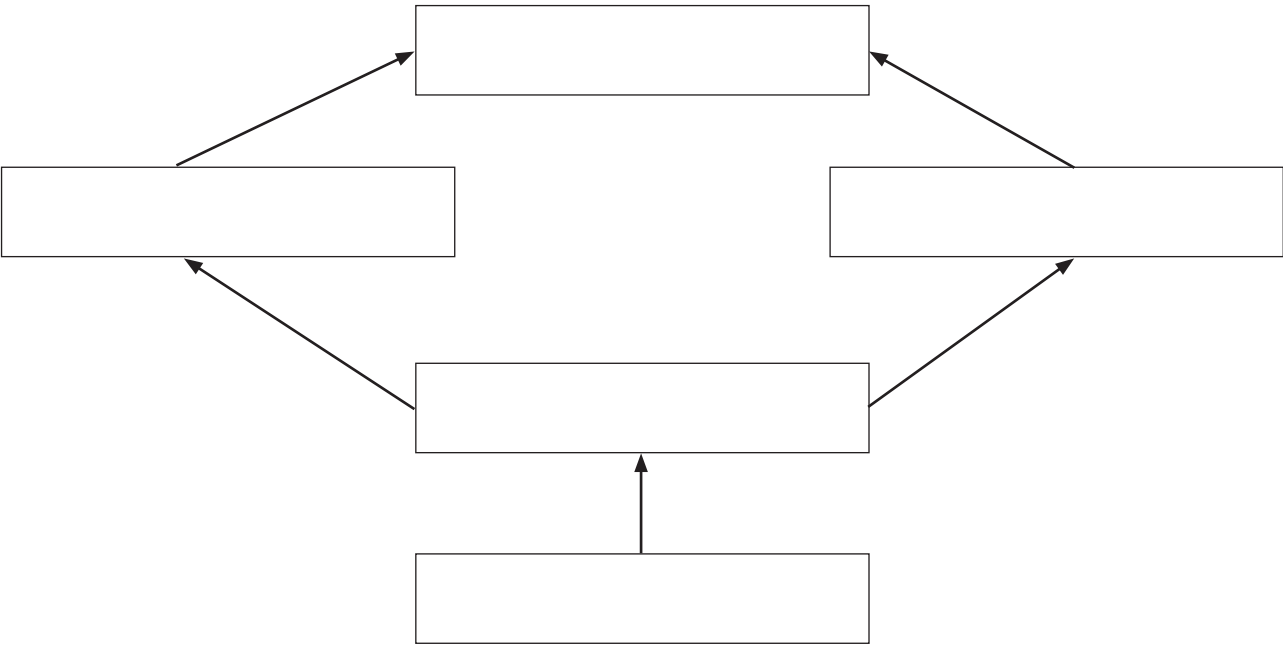
1. Killer whales (*Orcinus orca*) feed on thousands of large fish and squid. These large fish and squid feed on hundreds of small fish. Small fish feed on tiny organisms called plankton.

In 2016, a dead killer whale was found washed up on an island off the West Coast of Scotland. Tests revealed the whale contained one of the highest levels of PCBs (polychlorinated biphenyls) ever recorded. Levels of PCBs are measured in milligrams per kilogram of fat in an animal's body. Scientists believe that the threshold where PCBs can cause harm to killer whales is 30 mg/kg of fat. The level of PCB in the dead whale was 957 mg/kg of fat.

During the last century, PCBs were widely used in plastics, electrical goods and cement. PCBs were banned in the 1970s. It is estimated that there are a million tonnes of PCB-contaminated material waiting to be disposed of in Europe. PCBs are released into the environment from decomposing products that have not been disposed of properly. Plankton absorb PCBs that are washed into the oceans.

There are only eight resident killer whales remaining in UK waters. Scientists have not observed any offspring being born to the resident killer whales in the 25 years they have been studying them.

- (a) (i) Complete the simple food web below to show the feeding relationships of a killer whale. [3]



- (ii) State the term used to describe the feeding stage of killer whales. [1]

.....



Examiner
only

- (b) Calculate how many times greater the level of PCBs was in the dead killer whale’s body when compared to the threshold where PCBs cause harm. [1]

Times greater =

- (c) Suggest the name of a group of micro-organisms responsible for releasing PCBs into the environment. [1]

.....

- (d) Explain how the PCBs led to the death of the killer whale. [2]

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- (e) Apart from causing death, describe another effect of PCBs on killer whales in UK waters. Give evidence to support your answer. [2]

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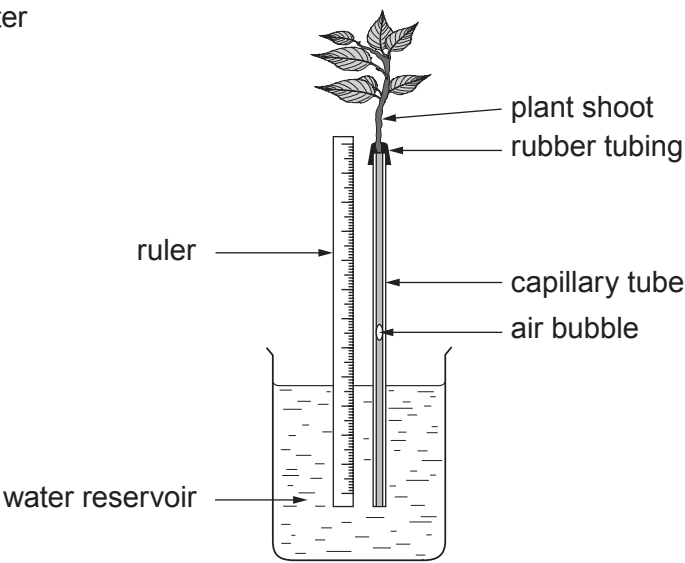
- (f) Suggest why PCBs are still present in UK waters even though they have been banned since the 1970s. [1]

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2. Students used a simple potometer as shown below to compare the rate of transpiration in shoots from a beech tree (*Fagus sylvatica*) and a bay tree (*Laurus nobilis*).

A simple potometer



(a) State the name of the tissue that transports water in plants. [1]

.....

(b) State **two** uses of water in plants. [2]

.....
.....

(c) (i) When the beech tree shoot was used, the air bubble in the capillary tube took 87 seconds to travel 50 mm. Calculate the rate of transpiration for the shoot from the beech tree in mm/s. [1]

Rate of transpiration = mm/s

(ii) When the bay tree shoot was used, the air bubble in the capillary tube took 51 minutes and 58 seconds to travel 50 mm.

State **one** conclusion that could be reached from the results. [1]

.....
.....



Examiner
only

(d) (i) State **one** factor that should have been kept constant in this investigation. [1]

.....

(ii) Describe how the students could improve their confidence in the results. [1]

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(e) Explain why a potometer cannot be used to study transport of sugars in a plant. [2]

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3. A student examined cells from an onion epidermis using a light microscope.

The student used the following method:

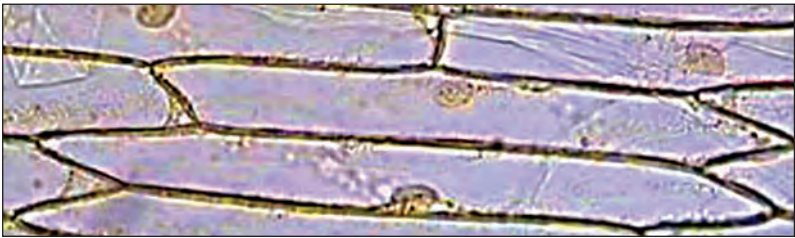
- 1. Peel a thin layer of epidermis from inside a freshly cut onion.
- 2. Lay epidermis on glass slide.
- 3. Add a drop of iodine solution to the onion epidermis on the glass slide.
- 4. Gently lower a coverslip onto the glass slide.
- 5. Using a light microscope examine the slide to identify some of the cell structures.

(a) One hazard for this method has been assessed and given in the table below. Identify **one** further hazard involved in this method and complete the risk assessment for it in the table below. [2]

Hazard	Risk	Control measure
Iodine solution is an irritant	Transfer of iodine solution from hand to eye can irritate the eyes	Use lowest concentration possible
.....		
.....		



(b) A photograph of the student’s observation is shown below.



(i) Draw **one** of the onion cells from the photograph above in the space below. [1]

(ii) **On your drawing**, label **three** parts of the cell **you can see** in the photograph. [2]

(c) State how the student would calculate the total magnification of the microscope she was using. [1]

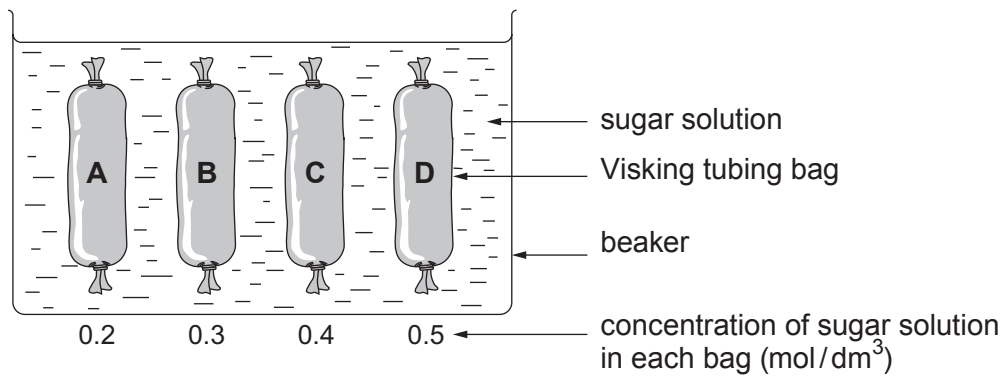
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4. A group of students carried out the following investigation.
- Four identical Visking tubing bags were filled with 10 cm³ of a sugar solution.
 - Each bag contained a different concentration of sugar solution.
 - The outsides of the bags were washed and dried and then the mass of each bag was recorded.
 - The bags were placed in a large beaker containing a sugar solution of unknown concentration.
 - The bags were left for 30 minutes.
 - The bags were removed from the beaker, the outsides of the bags were dried and their final mass recorded.
 - The percentage change in mass was calculated.



The results are shown in the table.

Visking tubing bag	Concentration of sugar in the Visking tubing bag (mol/dm ³)	Percentage change in mass at 30 minutes (%)
A	0.2	-2.6
B	0.3	+1.3
C	0.4	+3.6
D	0.5	+4.2



Examiner
only

(a) Explain the result for Visking tubing bag **D**.

[4]

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(b) Estimate the concentration of the sugar solution in the beaker. Give a reason for your answer.

[2]

Estimated concentration = mol/dm³

Reason:

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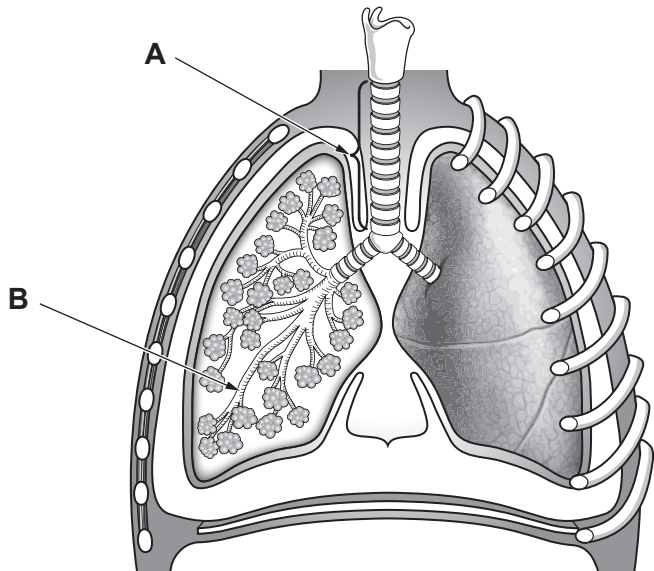
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5. The diagram shows a section through the human thorax.



(a) State the names of structures **A** and **B**.

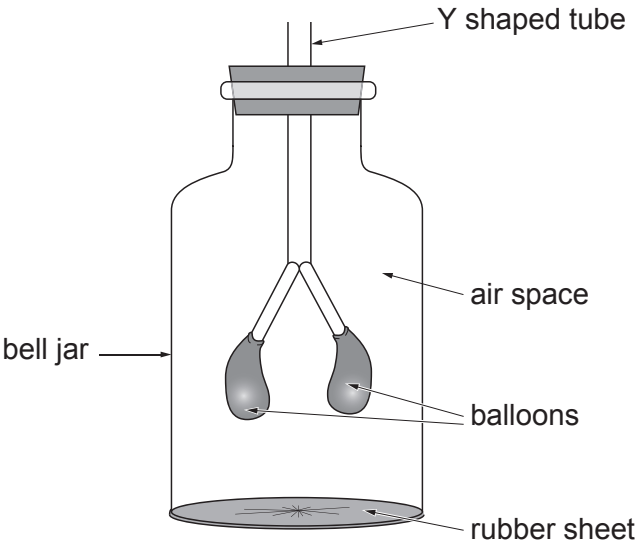
[2]

(i) **A**

(ii) **B**



A bell jar model can be used to demonstrate inspiration and expiration.



(b) Describe and explain how the bell jar model could be used to demonstrate **inspiration**. [4]

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(c) Describe **two** limitations of the bell jar as a model to demonstrate inspiration. [2]

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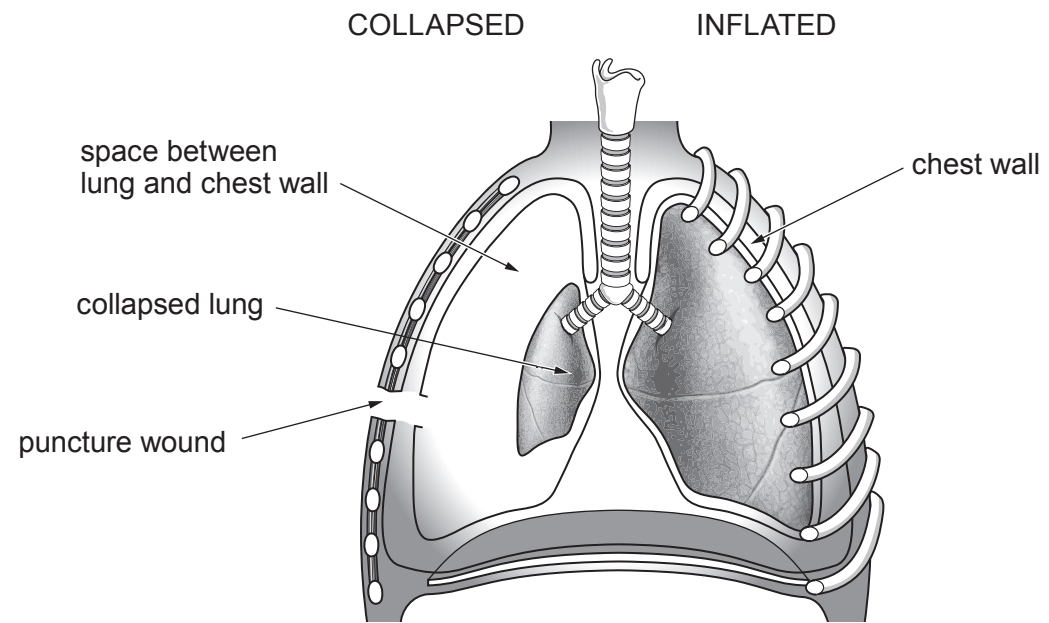
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- (d) A collapsed lung, as shown below, can be caused by an injury to the chest, such as a broken rib or puncture wound. Air enters the space between the lung and the chest wall causing the pressure in the thorax to increase and the lung to collapse.



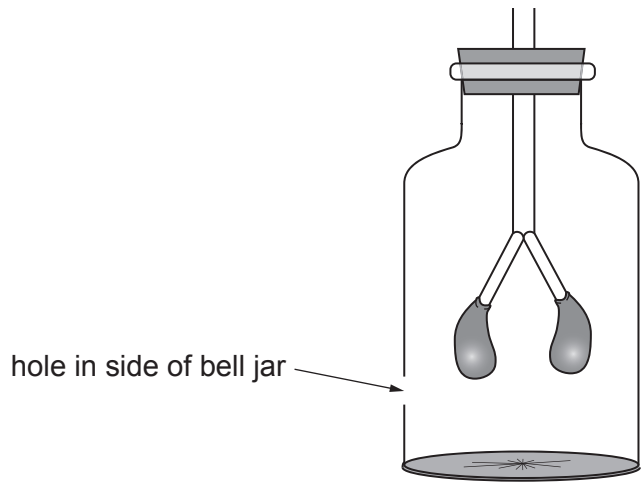
- (i) Suggest why a collapsed lung would not be considered to be life-threatening. [1]

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Some students attempted to model a punctured lung using a bell jar model.



(ii) Describe and explain how the hole will affect the bell jar model when the rubber sheet is pulled down. [2]

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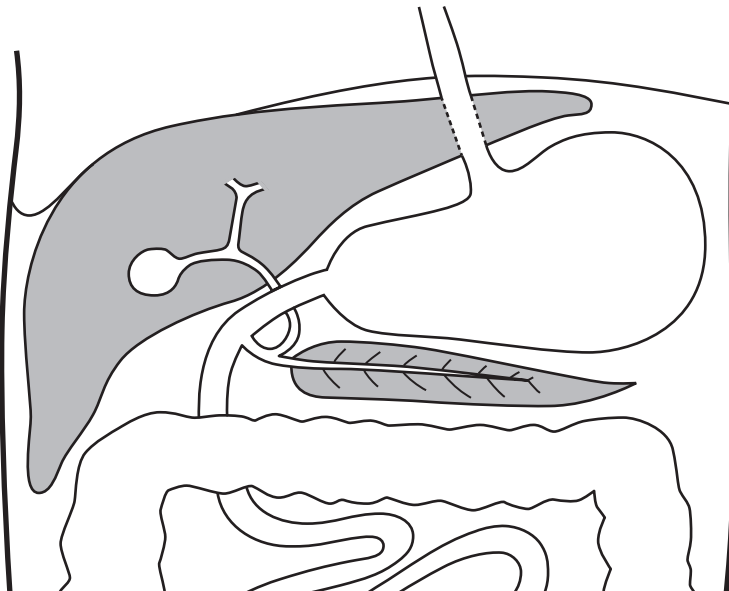
6. Cystic fibrosis is a genetic condition that affects many organs in the body including the pancreas. People with cystic fibrosis may have blockages in the pancreas which prevent enzymes from the pancreas reaching the small intestine. To overcome the problem of these blockages, people with cystic fibrosis take pancreatin tablets.

Pancreatin tablets have a tough coating that can resist low pH values. They contain a mixture of digestive enzymes normally produced in the pancreas.

- (a) State the name of the process that moves the tablet along the digestive system after it has been swallowed. [1]

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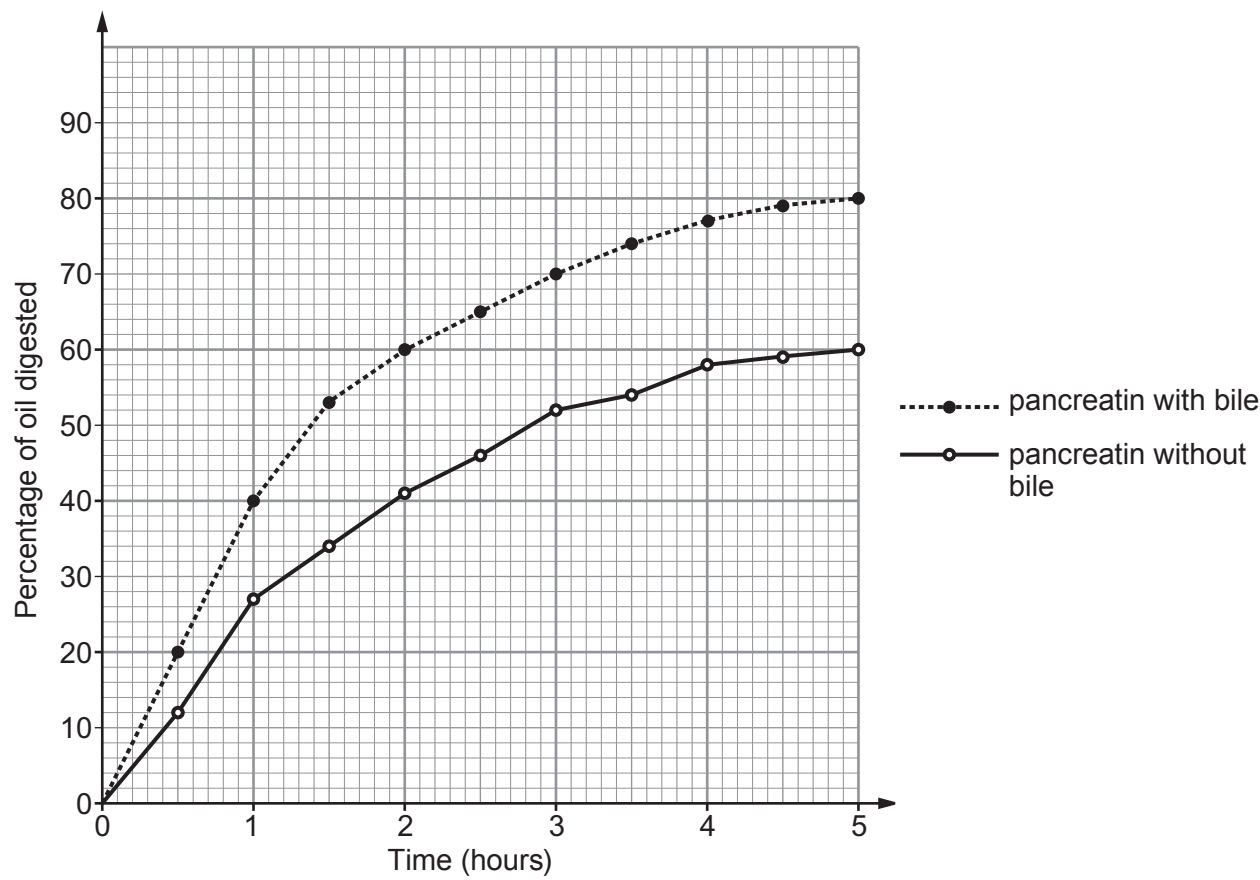
- (b) The diagram below shows part of the digestive system. [2]



- (i) Use a line labelled **X** to suggest the location of the blockage caused by cystic fibrosis.
- (ii) Use a line labelled **Y** to show the organ that secretes bile.



(c) The graphs below show the results of some experiments involving pancreatin.



Graph – Percentage of olive oil digested by pancreatin with and without bile.

(i) Pancreatin contains three different digestive enzymes, one of which is lipase. State the name of the *other two* enzymes present in pancreatin. [2]

I.

II.

(ii) Explain the effect of bile on the digestion of olive oil. [3]

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Examiner
only

(iii) State the products formed when olive oil is digested and give **one** use of these products in the body. [2]

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(d) Suggest why pancreatin tablets need to have ‘a tough coating that can resist low pH values’. Explain your answer. [2]

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(e) State the name of the group of molecules to which enzymes belong and explain why the three enzymes in the pancreatin tablets have different structures. [2]

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14



7. The following extract is from an article published on the BBC news website.

Mega chicken factory plan approved

Campaigners have lost their battle against plans for a huge poultry factory after the proposal was approved by councillors.

The farm will house nearly 180 000 chickens.

Villagers say the factory will cause unacceptable levels of noise, smell and traffic.

The company behind the scheme insists it will improve the local environment.

Outline the advantages and disadvantages of allowing the development of the chicken farm and suggest why an environmental impact assessment would be required. [6 QER]

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8. The picture shows a label from a packet of pasta.

Typical values (as consumed)	per 100g
Energy	658kJ
	155Kcal
Fat	0.6g
of which saturates	0.1g
Carbohydrate	30.3g
of which sugars	0.4g
Fibre	3.4g
Protein	5.5g
Salt	trace

(a) From the label above, identify the nutrient that provides bulk in the diet. [1]

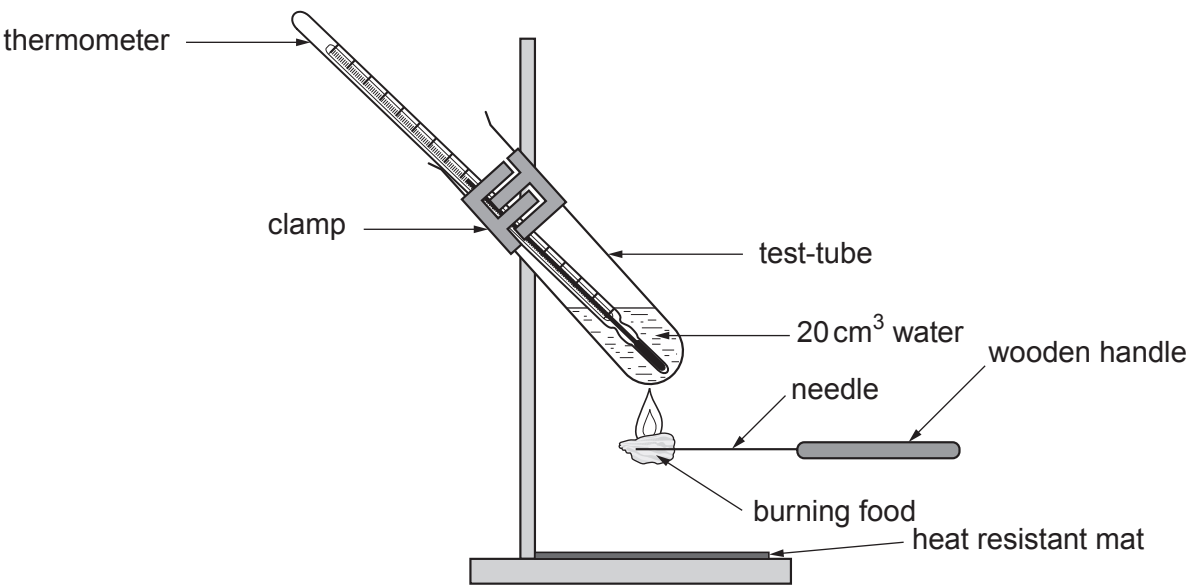
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(b) State why the government recommends that added sugars should not make up more than 5% of the energy obtained from food and drink each day. [1]

.....



Students compared the energy content of crisps and pasta using the following apparatus.



The energy content of food can be calculated using the following formula.

$$\text{Energy (J/g)} = \frac{(4.2 \times \text{volume of water} \times \text{change in temperature})}{\text{mass of food}}$$

The following results were obtained.

Table 1

Food	Mass of food (g)	Temperature of water in test tube (°C)		Change in temperature (°C)
		Initial	Final	
Crisp	0.7	20	59	39
Pasta	1.3	20	36	16



Examiner
only

The table below shows the energy values for the crisps.

Table 2

Food	Energy calculated from the experiment (J/g)	Energy calculated from the experiment (kJ/g)	Energy recorded on the food label (kJ/g)
Crisp	4680	4.680	22.788
Pasta			

- (c) Complete **Table 2** to show the energy values for pasta. You will need to refer to **Table 1** opposite and the food label on page 19. [4]

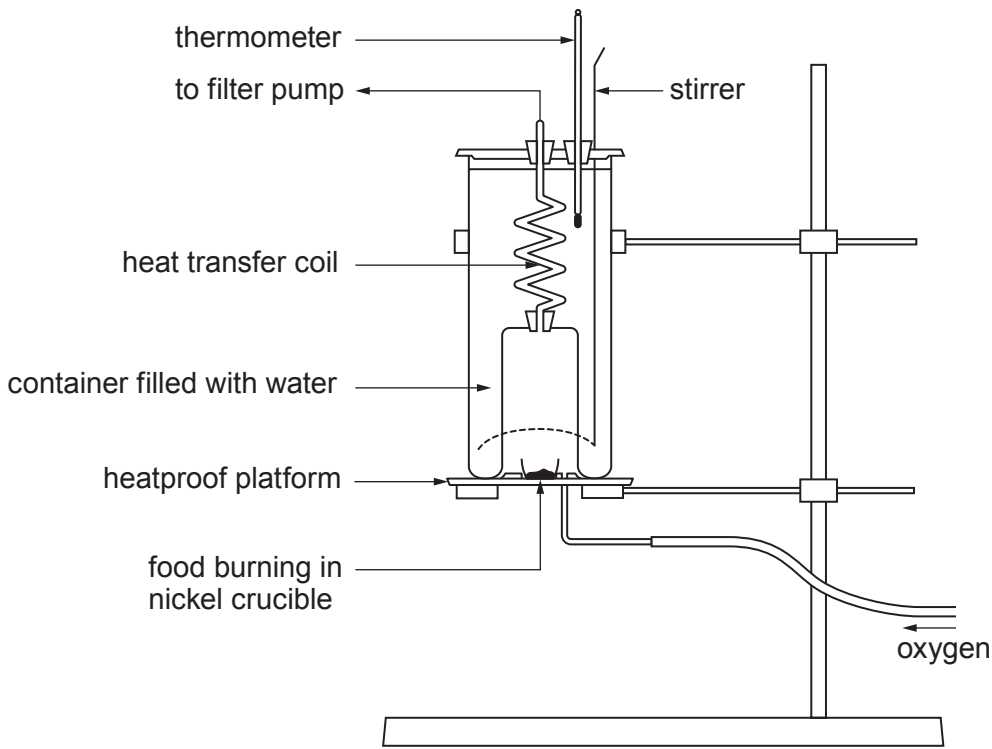
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- (d) Use the information from **Table 2** to calculate the percentage by which the value on the label for the crisps was greater than the value obtained in the experiment. [2]

Answer = %



- (e) The value calculated on the food label is worked out experimentally in a laboratory using apparatus similar to that shown below.



Suggest **two** ways in which the design of this apparatus makes it more efficient than the class experiment. [2]

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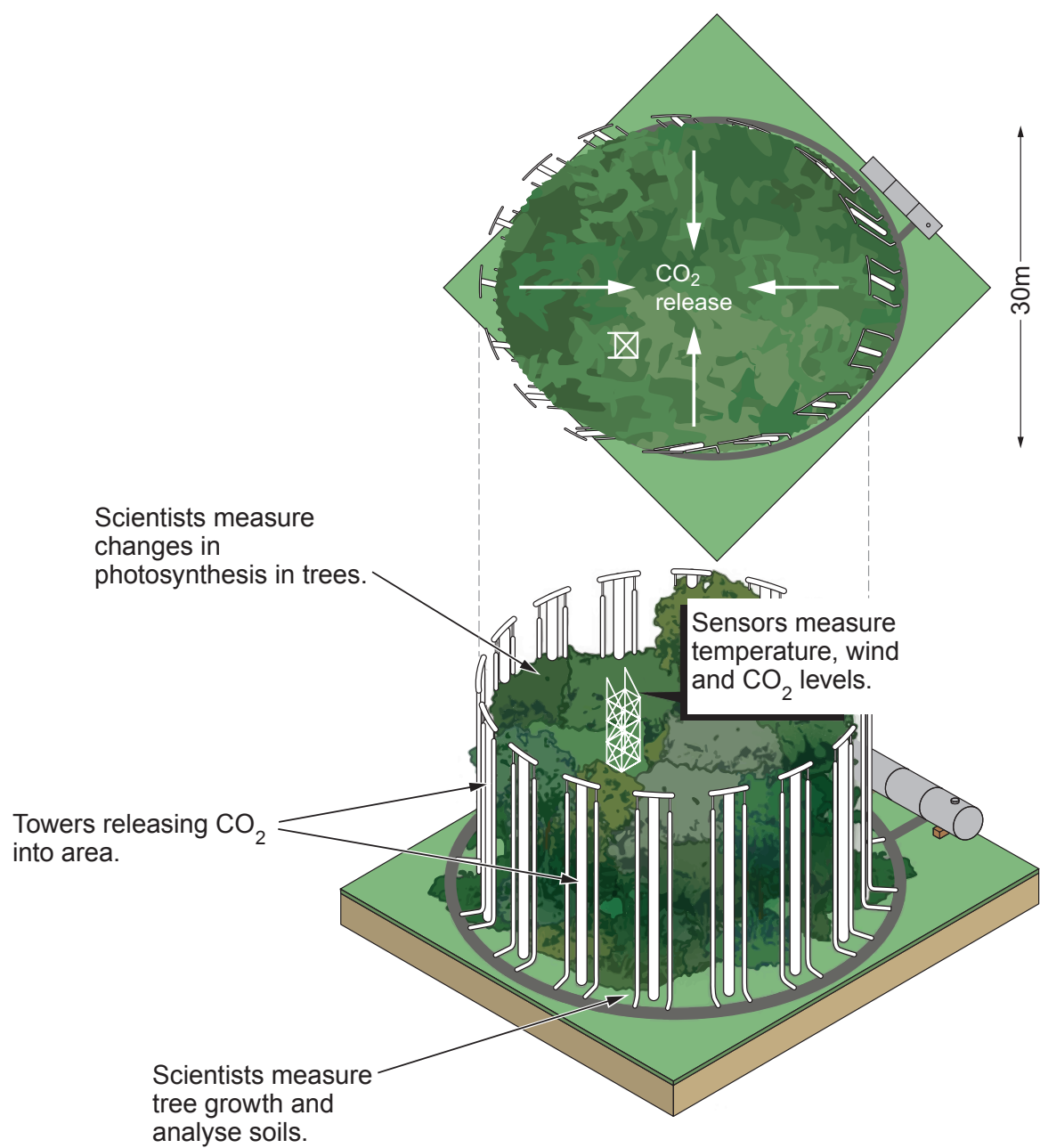


9. The concentration of carbon dioxide (CO₂) in the atmosphere has risen from a mean of 315 parts per million (ppm) in 1959 to a current mean of 385 ppm. It is predicted that the concentration of CO₂ will continue to rise to between 500 and 1000 ppm by the year 2100.

Most of our current knowledge on plant responses to CO₂ concentrations in the atmosphere is based on experiments carried out in greenhouses.

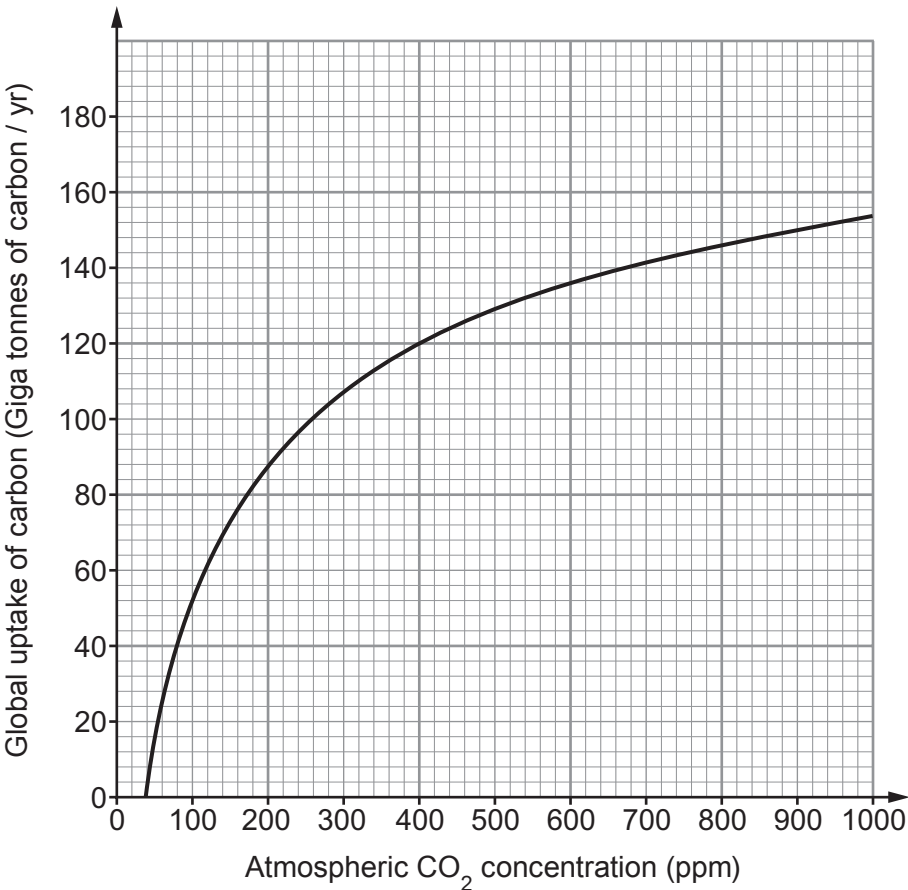
Free-air carbon dioxide enrichment (FACE) experiments have been carried out around the world to study the effects of increased CO₂ levels on photosynthesis in plants outside greenhouses. This will enable scientists to make conclusions about the effect of increased concentrations of CO₂ in the future.

Figure 1 – Illustration of a FACE experiment – aerial view (above), side view (below)



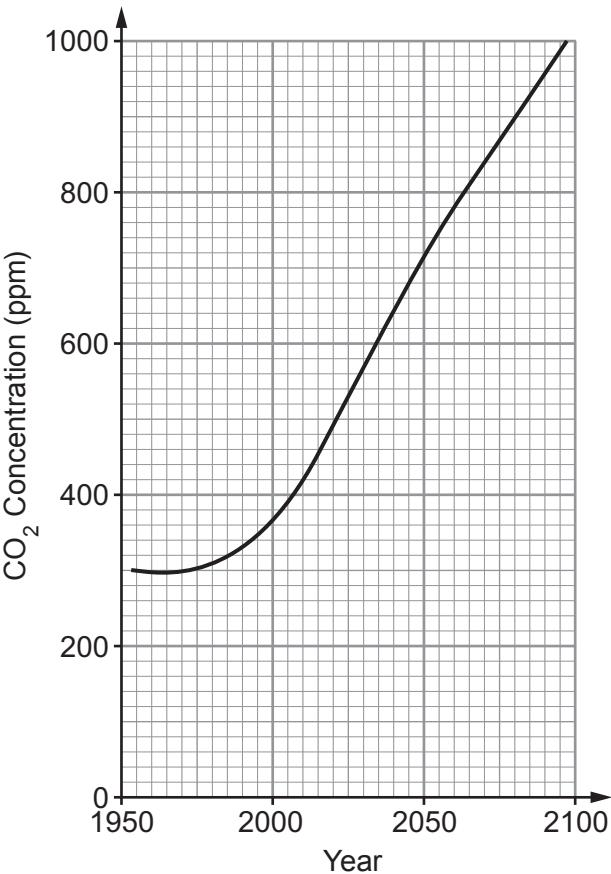
Scientists have used the results of many FACE experiments to predict the mass of CO₂ taken in by plants each year. **Graph 1** shows the effect of atmospheric CO₂ concentration on annual global uptake of carbon.

Graph 1



The International Panel on Climate Change (IPCC) uses many different computer models to predict how CO₂ concentration in the atmosphere will change over time. The results of one of these models is shown in **Graph 2**.

Graph 2



Examiner
only

(a) (i) Write the word equation for photosynthesis. [2]

.....
(ii) State the name of **one** molecule made in plants from glucose. [1]

.....

(b) (i) Using both **Graphs 1** and **2**, estimate the predicted global uptake of carbon in 2080. [1]

.....
(ii) Suggest an explanation for the shape of **Graph 1** between 400 and 800 ppm. [1]

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.....

(c) (i) State how scientists increased the validity of their conclusions about the global changes to CO₂ concentrations in the atmosphere. [1]

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(ii) Suggest why it is important that scientists use FACE experiments on plant growth to study the effects of increasing CO₂ concentrations instead of similar experiments in greenhouses. [1]

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END OF PAPER

7



Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3400UA0-1



Z22-3400UA0-1

WEDNESDAY, 15 JUNE 2022 – MORNING

BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	9	
4.	5	
5.	12	
6.	11	
7.	8	
8.	7	
9.	8	
Total	80	

3400UA01
01**ADDITIONAL MATERIALS**

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6(b)** is a quality of extended response (QER) question where your writing skills will be assessed.

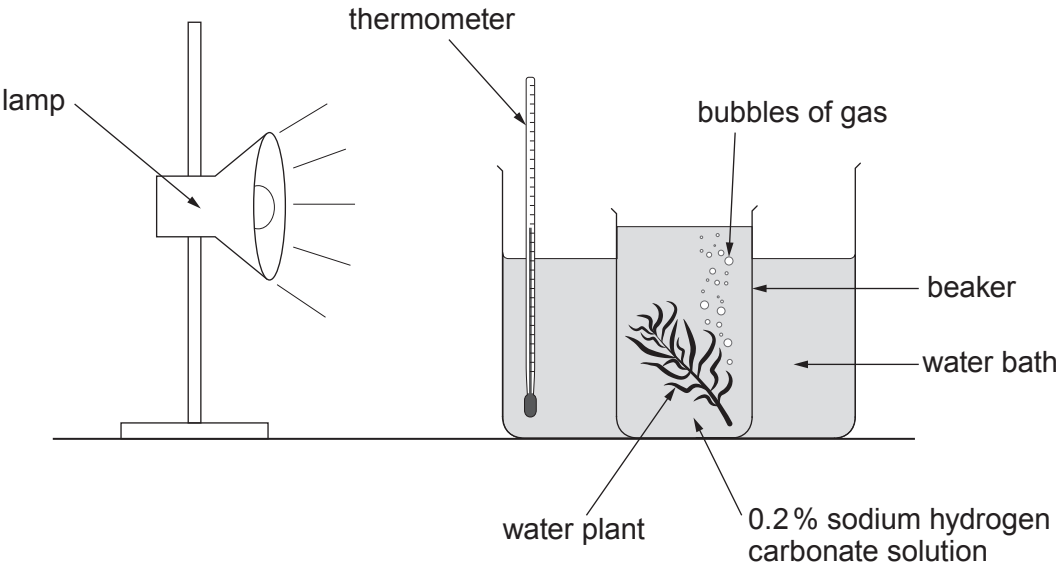


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Answer **all** questions.

1. Students were instructed to investigate the effect of temperature on photosynthesis. They used the apparatus shown in **Image 1.1**.

Image 1.1



- The number of bubbles of gas produced by the water plant per minute was counted at different temperatures.
- The experiment was carried out three times at each temperature.
- The results are in **Table 1.2**. Means were calculated to the nearest whole number.

Table 1.2

Temperature of water bath (°C)	Number of bubbles of gas produced per minute			
	Test 1	Test 2	Test 3	Mean
20	18	21	12	17
25	15	24	21	20
30	24	27	30	27
35	26	25	26	26
40	14	13	10	12
45	3	5	6	
50	0	0	0	0



Examiner
only

(a) (i) State the name of the gas produced during photosynthesis. [1]

.....

(ii) State **two** ways in which plants use the glucose produced during photosynthesis. [2]

I.

II.

(b) (i) Calculate the mean number of bubbles per minute for 45 °C. **Write your answer in Table 1.2.** [2]

(ii) Describe the relationship between the temperature and the number of bubbles produced per minute. [2]

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(iii) Explain the result for 50 °C. [2]

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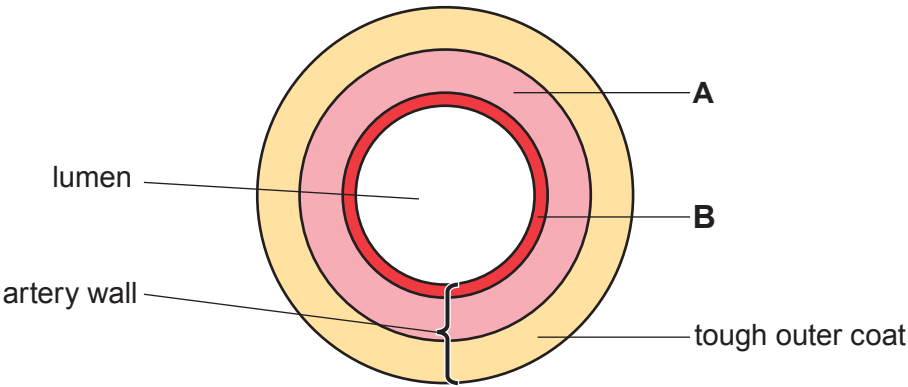
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2. Image 2.1 shows a cross section through a healthy coronary artery.

Image 2.1



(a) (i) State the names of **tissues A** and **B**. [2]

A

B

(ii) Suggest how the diameter of the lumen would differ in an individual with cardiovascular disease and explain the difference. [2]

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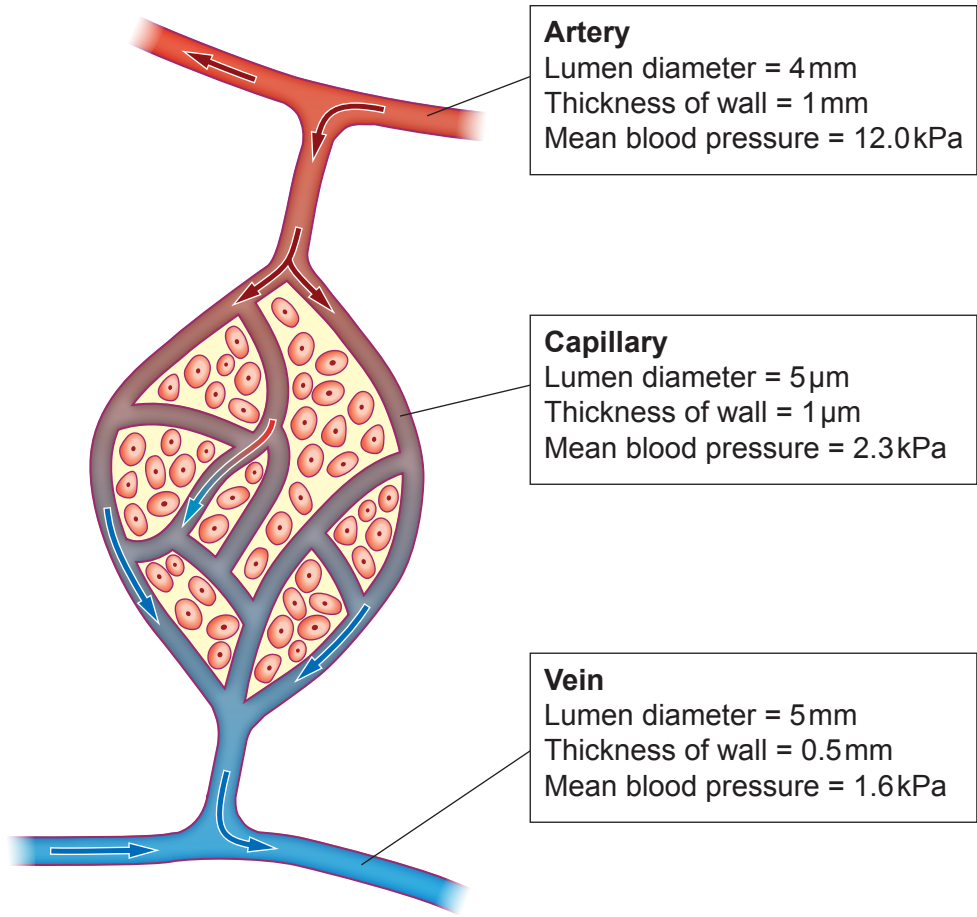
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(b) **Image 2.2** shows a cross section of blood vessels found in muscle tissue, along with some information about them.

Image 2.2



(i) Use **Image 2.2** to give **two** differences between the structure of an artery and a vein. [2]

.....

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(ii) Veins contain structures that are not present in arteries or capillaries. State the name of these structures and describe their function. [2]

Name of structures

Function

.....

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Examiner
only

(c) (i) State the name of **one** substance which passes from the blood into the body cells. [1]
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(ii) State **one** way in which capillaries are adapted for their function. Give an advantage for this adaptation. [2]

Adaptation

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Advantage

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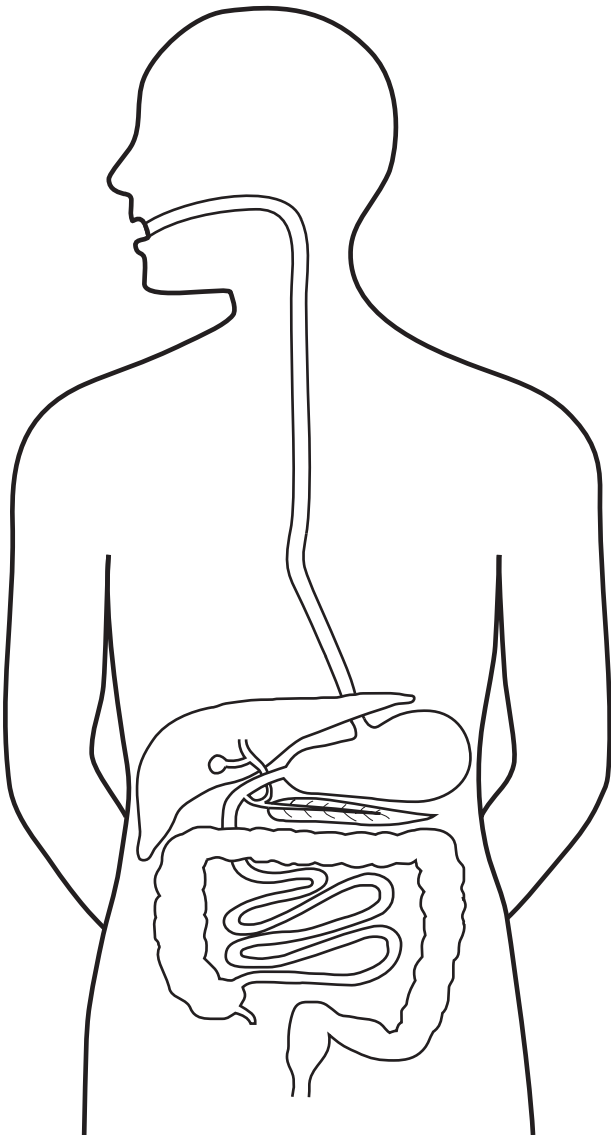
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3. (a) **Image 3.1** shows the human digestive system.

- (i) **Draw an arrow** labelled **P** on **Image 3.1** to show an organ where **only** protein is digested. [1]
- (ii) **Draw an arrow** labelled **B** to show an organ where fat and oil (lipids) are digested. [1]

Image 3.1



- (iii) State the purpose of digestion. [2]

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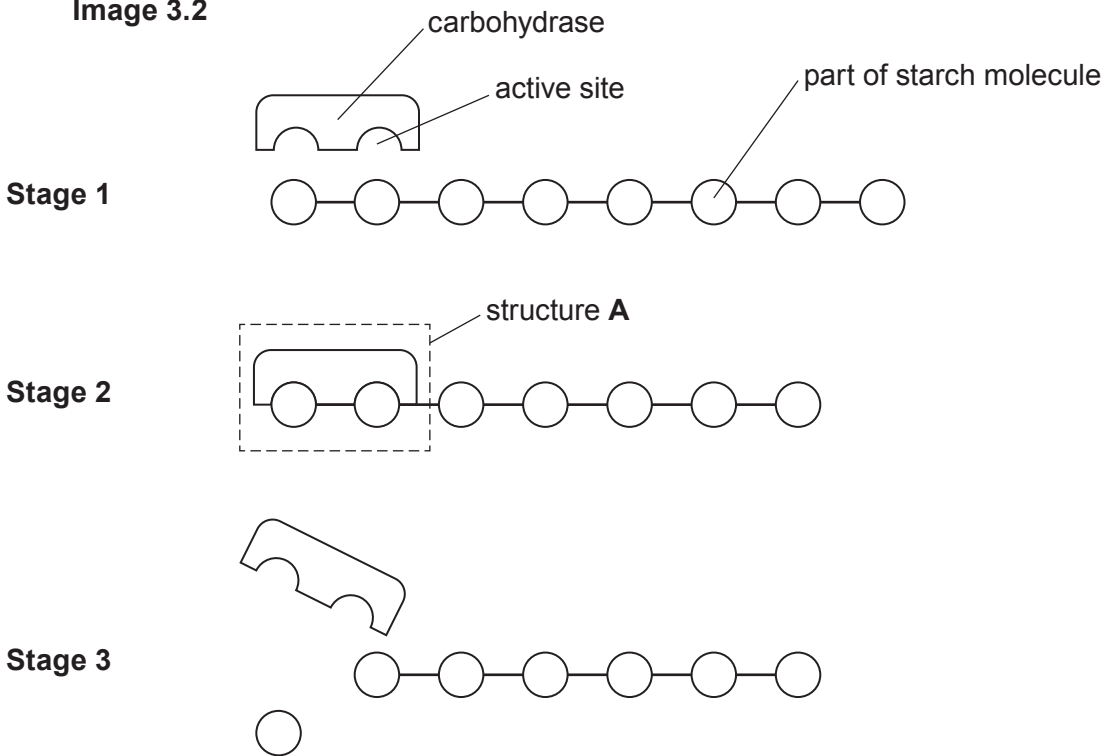
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(b) **Image 3.2** shows part of a starch molecule being digested by carbohydrase in the mouth.

Image 3.2



(i) State the name of structure **A**, indicated by the box in **stage 2**. [1]

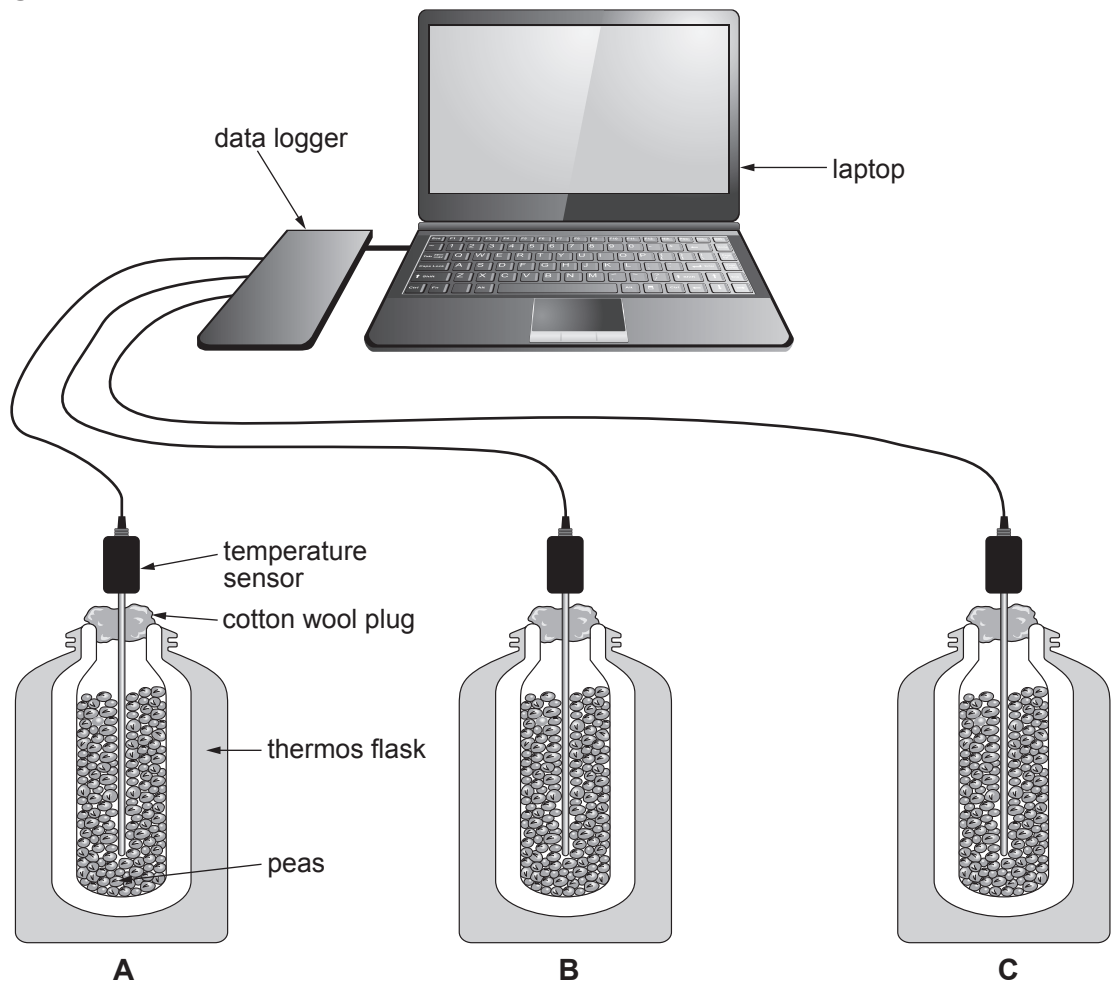
(ii) State the name of the molecule produced by the digestion of starch **and** describe **one** way in which it is used by the body. [2]

(iii) Describe and explain what will happen to the active site of the carbohydrase enzyme when it is inside the stomach. [2]



4. The experiment shown in **Image 4.1** was set up to investigate respiration in germinating peas.

Image 4.1



The flasks all contained an equal mass of peas. **Table 4.2** shows the treatment that each flask received:

Table 4.2

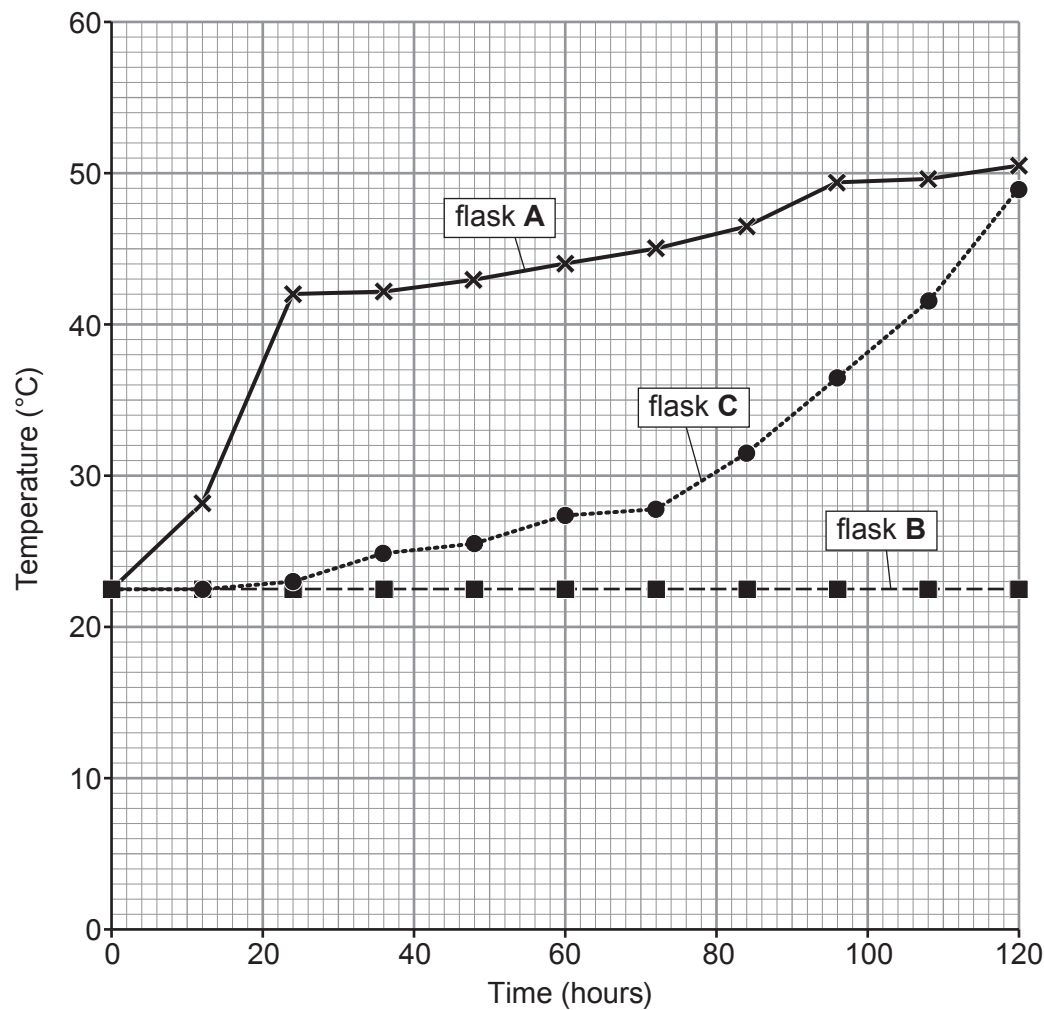
Flask	Treatment
A	Germinating peas soaked in disinfectant.
B	Germinating peas which had been boiled for 30 minutes then soaked in disinfectant.
C	Germinating peas which had been boiled for 30 minutes.

The temperature of the flasks was monitored for 120 hours.



The results of the experiment are shown below in **Graph 4.3**.

Graph 4.3



- (a) (i) State the purpose of soaking the peas in disinfectant in flasks **A** and **B**. [1]

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.....

- (ii) Explain the results for flask **A** between 0 and 24 hours. [2]

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Examiner
only

(b) Explain how flask **B** acts as a control experiment for flask **A**. [1]

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(c) Aerobic respiration of one molecule of glucose releases 38 molecules of ATP. State whether the number of molecules of ATP produced from anaerobic respiration would be **higher** or **lower** than 38. Give a reason for your answer. [1]

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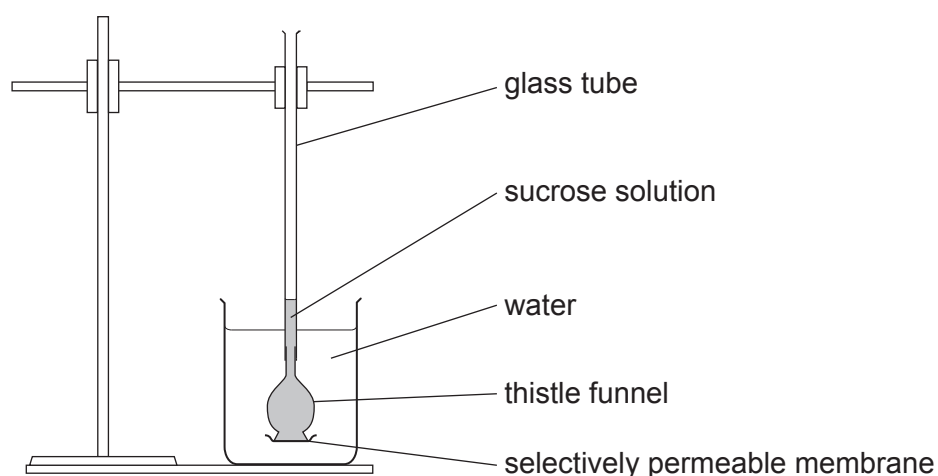
5



5. The following experiment was used to study the movement of water in and out of cells.

- A glass thistle funnel was half filled with sucrose solution.
- The bottom of the thistle funnel was closed by a selectively permeable membrane.
- The thistle funnel was placed in a beaker containing distilled water. As shown in **Image 5.1**.
- The level of sucrose solution in the glass tube was marked.
- The apparatus was left for 30 minutes and then the level of the sucrose solution in the glass tube was measured.
- The distance the sucrose solution had moved was calculated.
- If the level of the sucrose solution went up, the number was recorded as a positive number, but if the level of the sucrose solution went down, the number was recorded as a negative number.
- The experiment was repeated by placing the thistle funnel in salt solutions of 0.2, 0.3, 0.6 and 0.8 mol/dm³.

Image 5.1



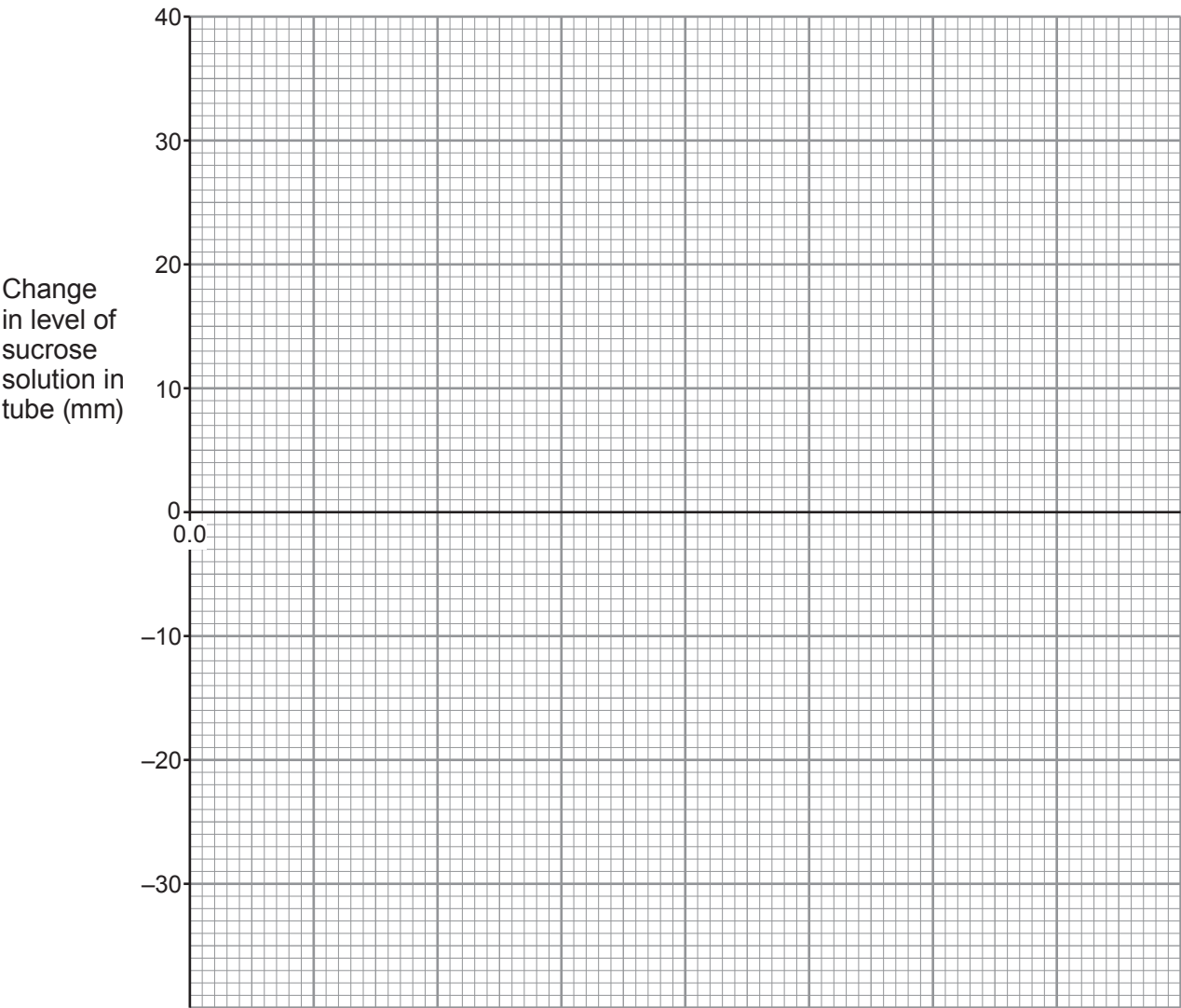
The results of the experiment are shown in **Table 5.2**.

Table 5.2

Salt concentration (mol/dm ³)	Change in level of the sucrose solution in tube (mm)
0.0 (distilled water)	+36
0.2	+23
0.3	−7
0.6	−17
0.8	−23

- (a) Use the results from **Table 5.2** to produce a line graph on **Graph 5.3** below. The scale and label for the y axis have been completed for you, along with the origin for the x axis [5]

Graph 5.3



Examiner
only

- (b) (i) Explain why the sucrose solution moved up the glass tube when the thistle funnel was placed in distilled water. [4]

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- (ii) Use **Graph 5.3** to estimate the concentration of the sucrose solution in the thistle funnel. [3]

Concentration mol/dm³

Explain your answer in terms of water movement at this concentration.

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12



6. Nitrate Vulnerable Zones (NVZ) are areas where nitrate concentrations are a risk to human health, or harmful to the aquatic environment.

Llangorse lake is the largest natural lake in south Wales. Approximately two-thirds of the surrounding land is used for intensive farming. Natural Resources Wales (NRW) has calculated that 99% of the nitrate found in Llangorse lake comes from intensive farming.

Table 6.1 is used by NRW to estimate the risk from nitrate levels in water.

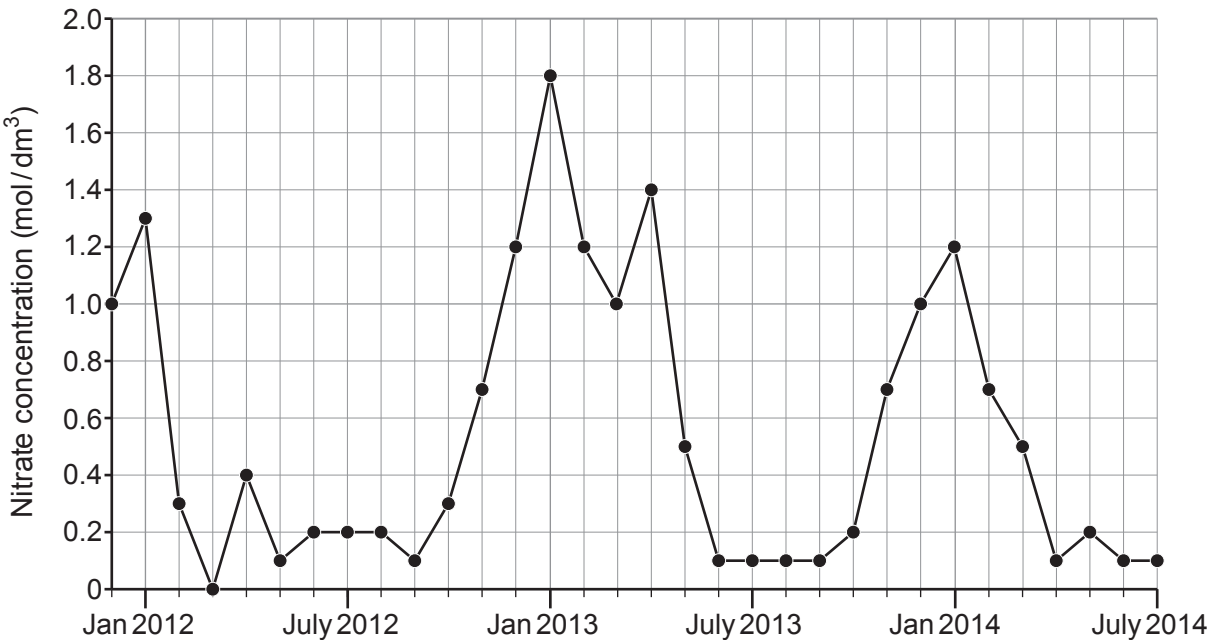
Table 6.1

Nitrate concentration (mol/dm ³)	Risk
< 0.5	very low
0.5 – 1.0	low
>1.0 – 1.7	low – medium
>1.7 – 3.5	medium
>3.5 – 6.0	medium – high
>6.0	high

Between December 2011 and July 2014, water samples were taken from Llangorse lake and the nitrate concentration analysed.

The results are shown in Graph 6.2.

Graph 6.2



Algal blooms were observed in Llangorse lake between April and October for each year shown in Graph 6.2.

- (a) (i) Using the information above, state the risk level for nitrates in Llangorse lake in January 2013.

[1]



Examiner
only

- (ii) Describe the annual trend common to all years shown in **Graph 6.2**. [2]

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- (iii) Suggest a possible explanation for the common annual decrease in the nitrate levels in Llangorse lake. [2]

- (b) Explain how intensive farming leads to the formation of algal blooms and how these algal blooms can be harmful to aquatic animals. [6 QER]

11

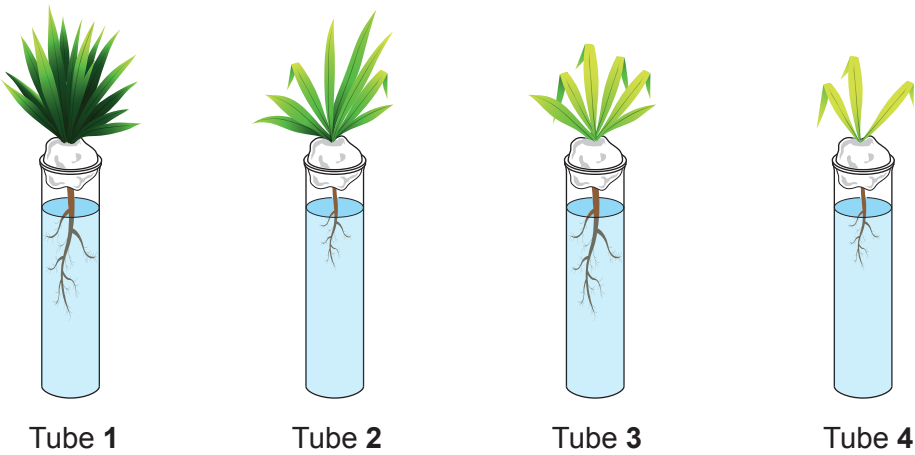


7. Julius von Sachs was a 19th century German scientist who studied the effect of plant nutrients on growth. He developed a growth solution which contained all the nutrients a plant needed for healthy growth.

A version of one of his experiments is shown below.

- 1. Seedlings of the same age were placed on cotton wool in four test tubes, each containing an equal volume of a different nutrient solution.
- 2. All four test tubes were placed on a shelf in full sunlight.
- 3. The growth of the seedlings was observed and recorded after one month (**Image 7.1**).

Image 7.1



	Tube 1	Tube 2	Tube 3	Tube 4
Content of growth solution	all nutrients present	no nutrient X	no nutrient Y	no nutrient Z
Description of growth after one month	healthy growth	poor root growth	yellowing of leaves	poor growth

(a) Identify the missing nutrients. [3]

Nutrient X =

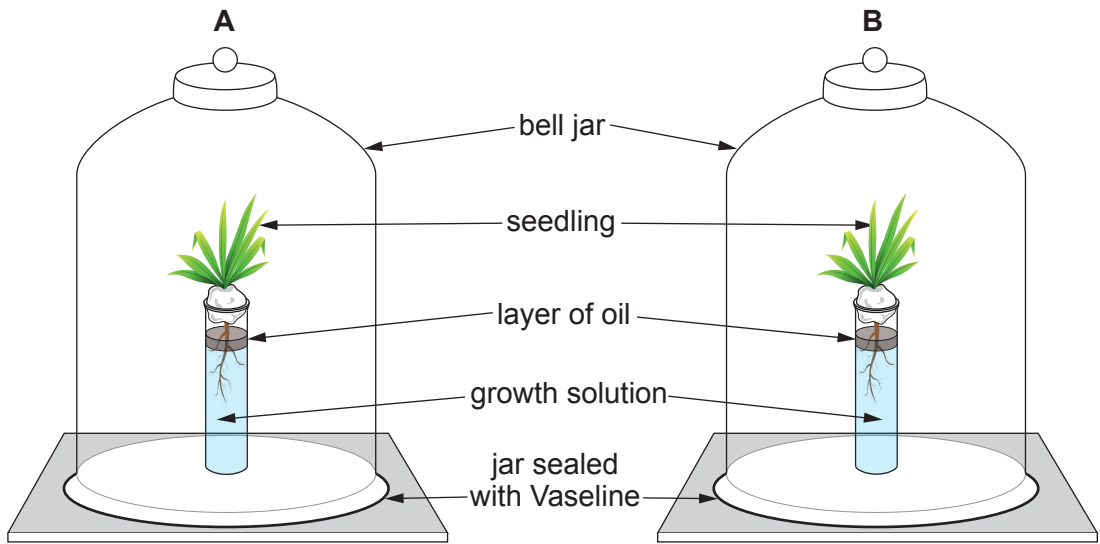
Nutrient Y =

Nutrient Z =



- (b) Students modified the experiment by taking two seedlings of similar age and placing them in a sealed bell jar as shown in **Image 7.2**.

Image 7.2



Bell jar A

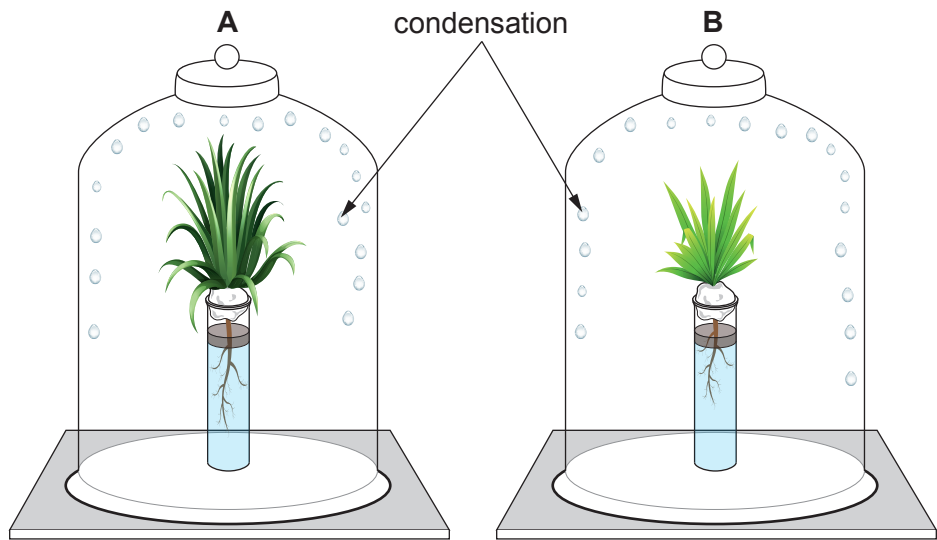
Seedling in growth solution containing all nutrients

Bell jar B

Seedling in growth solution containing all nutrients **and** a chemical which reduces the rate of respiration in root hair cells

Image 7.3 shows the appearance of the apparatus after 2 weeks.

Image 7.3



Examiner
only

(i) Suggest the purpose of the layer of oil in the experiment. [1]

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(ii) Explain the presence of the condensation (water droplets) inside the bell jars in **Image 7.3**. [2]

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(iii) Using your knowledge of how nutrients are absorbed, explain the difference in growth between the plants in bell jars **A** and **B** during the 2 weeks. [2]

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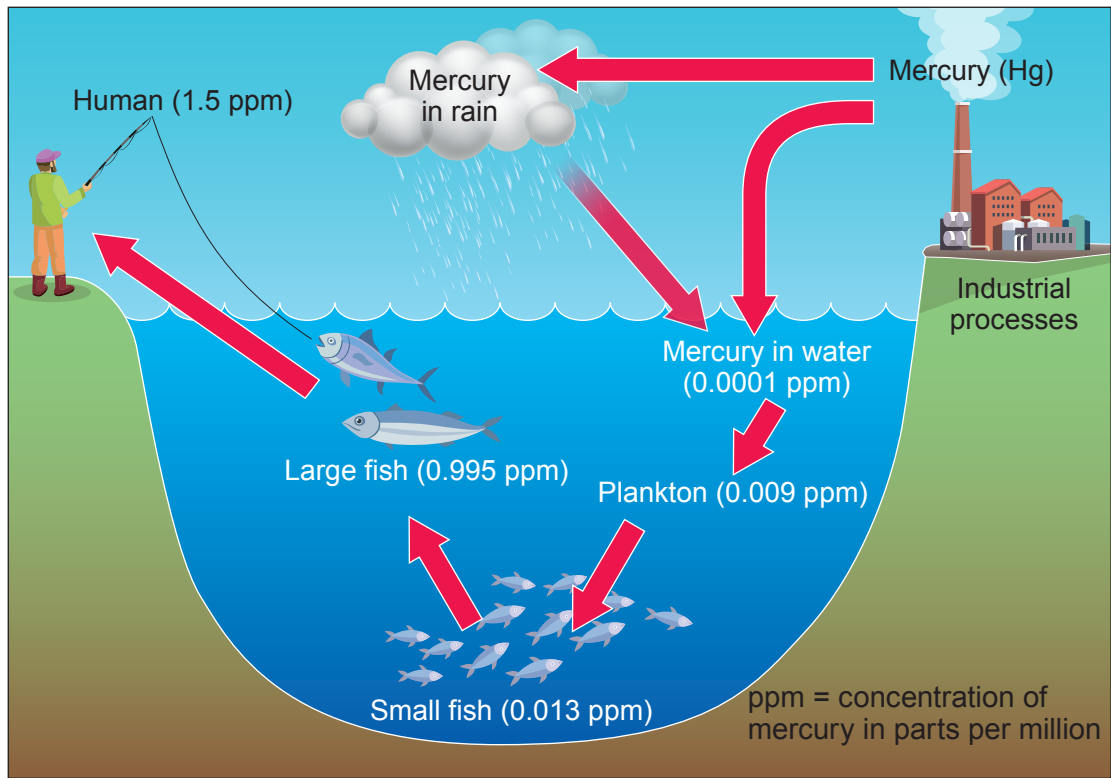
8



8. The United Nations Minamata Convention on Mercury is a global treaty to protect human health and the environment from the toxic effects of the heavy metal mercury.

In 2010, combustion of fossil fuels by industrial processes produced 24% of the total global emissions of mercury. **Image 8.1** shows how this mercury is cycled in the environment.

Image 8.1



- (a) (i) Calculate the percentage increase in mercury from the plankton to humans.
Give your answer to two significant figures.

[3]

Percentage increase =



Examiner
only

(ii) Using **Image 8.1** and your own knowledge, explain how the mercury released during the combustion of fossil fuels by industrial processes becomes harmful to humans. [3]

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(b) Apart from reducing mercury pollution, explain how reducing the combustion of fossil fuels can benefit the environment. [1]

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7



9. Popcorn lung is a lung disease which results in scarring and inflammation of the bronchioles. Symptoms of the disease are coughing and difficulty breathing. **Image 9.1a and 9.1b** show a healthy bronchiole and a bronchiole affected by popcorn lung.

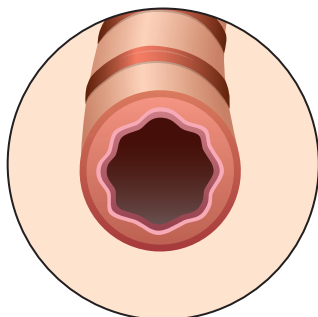


Image 9.1a – healthy bronchiole

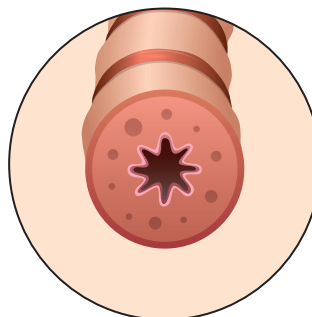


Image 9.1b – bronchiole affected by popcorn lung

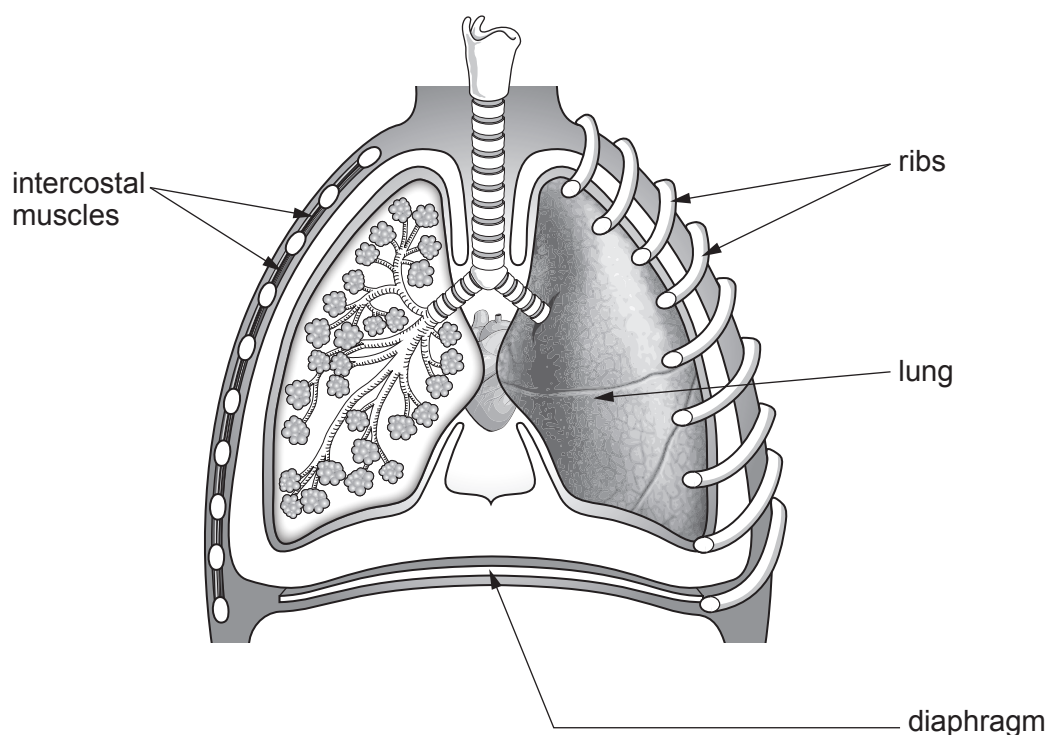
A chemical called diacetyl is used in the manufacture of popcorn. A link between inhaling diacetyl and the lung disease was made when a group of workers from a popcorn factory was found to have developed the disease.

In the US there have been increasing numbers of cases of popcorn lung linked to the use of e-cigarettes. There are around 7 700 flavours for e-cigarettes sold in the US. A single study of 51 e-cigarette flavours in 2016 found that 39 contained diacetyl. Diacetyl was banned in e-cigarette flavourings in the UK in 2016.

In 2019, a report in the journal *Nature* showed that diacetyl reduced the number of ciliated cells in the airways.

- (a) **Image 9.2** shows the human respiratory system.

Image 9.2



Examiner
only

(i) Use a labelled arrow to identify an alveolus on Image 9.2. [1]

(ii) State **two** ways in which the alveolus is adapted for gas exchange. [2]

.....

.....

(b) Explain why a person affected by popcorn lung shows the following symptoms:

(i) coughing. [1]

.....

.....

(ii) breathing difficulties. [2]

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(c) Use the information given to evaluate the validity of the following conclusion: [2]
'E-cigarette flavours cause popcorn lung'.

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END OF PAPER

8



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3400UA0-1



TUESDAY, 13 JUNE 2023 – MORNING

BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems
HIGHER TIER
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	12	
2.	8	
3.	8	
4.	8	
5.	9	
6.	5	
7.	7	
8.	8	
9.	15	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 8(b) is a quality of extended response (QER) question where your writing skills will be assessed.

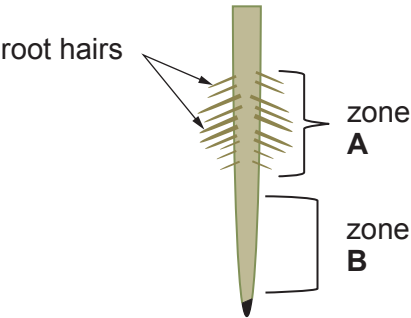


JUN233400UA0101

Answer **all** questions.

1. Part of a plant root is shown in **Image 1.1**.

Image 1.1



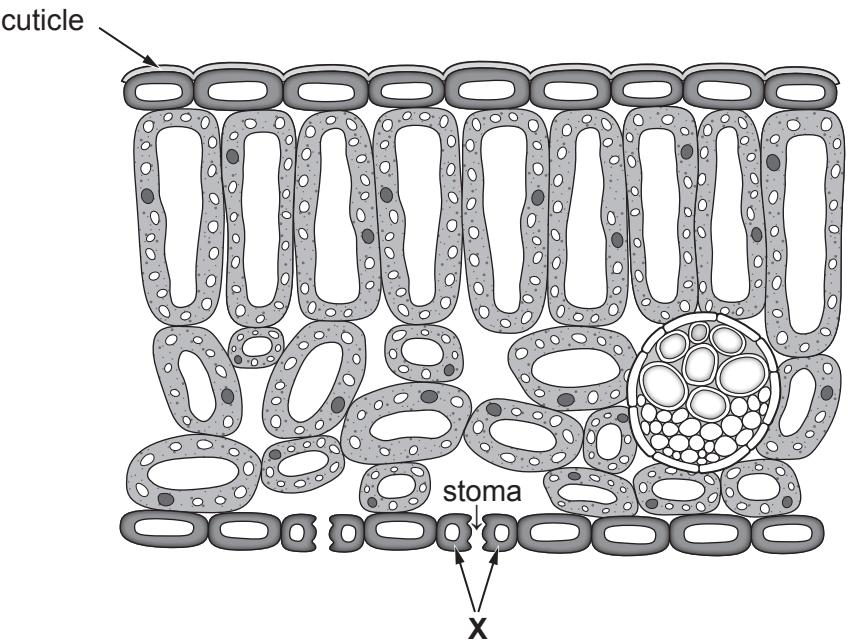
(a) Explain why water uptake in zone **A** is greater than in zone **B**. [1]

.....

.....

(b) **Image 1.2** represents a section through the leaf of a plant.

Image 1.2



(i) On **Image 1.2**, draw an arrow to show the tissue which transports water to all parts of the plant. Label the arrow with the name of the tissue. [2]



Examiner
only

(ii) I. Name cells **X** shown in **Image 1.2**. [1]

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II. State how the stoma and cuticle are involved in the control of water loss from a leaf. [2]

Stoma

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Cuticle

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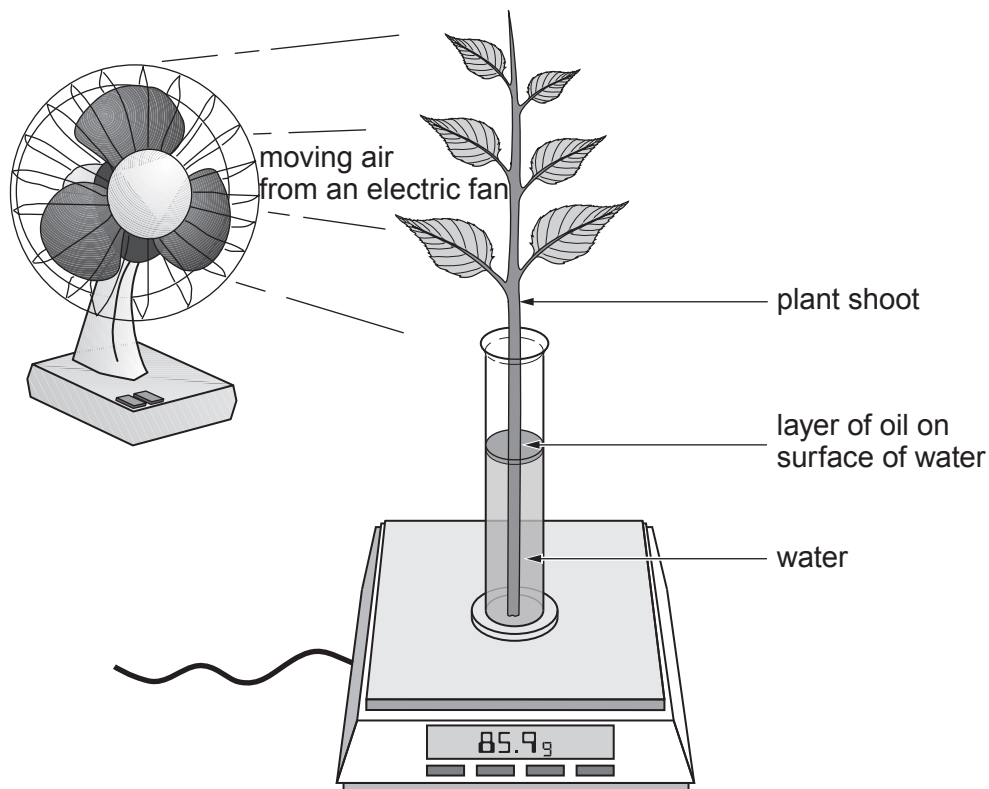
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03



- (c) Megan and Rhys investigated the loss of water from a leafy shoot. They used the apparatus shown in **Image 1.3**.

Image 1.3



They recorded the loss of mass after directing moving air, at different speeds, onto the shoot.

- (i) State the scientific term for the evaporation of water from the leaves of a plant. [1]

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- (ii) State **one** way in which Megan and Rhys could ensure that they carried out a fair test. [1]

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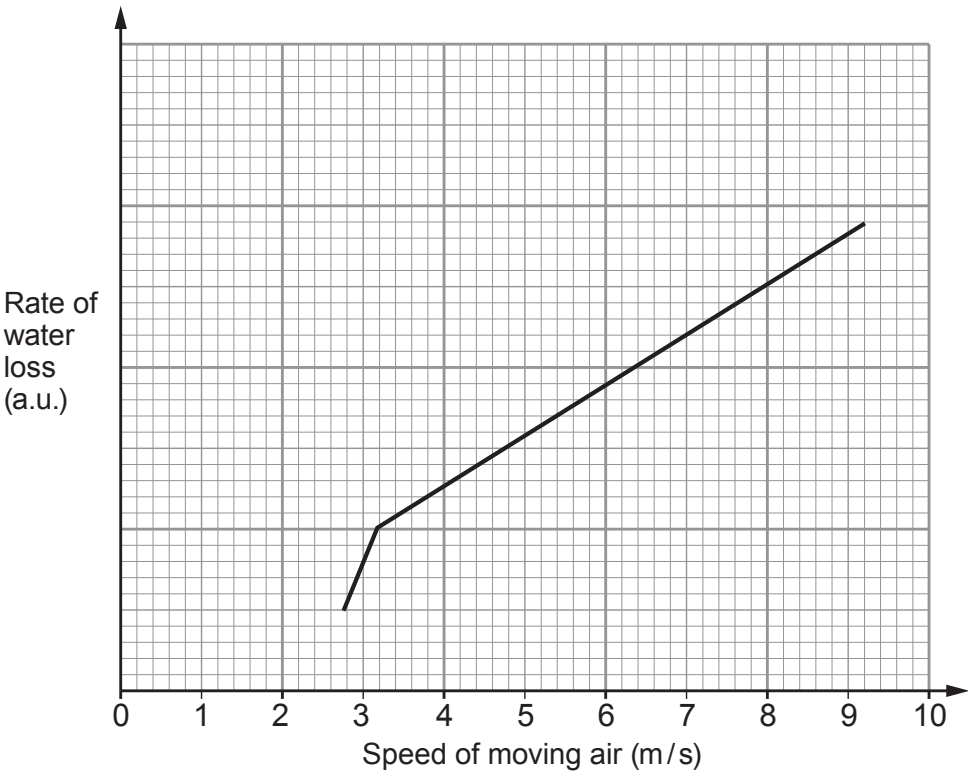
- (iii) State why it was important that the layer of oil was added. [1]

.....



(iv) The results of their investigation are summarised in **Graph 1.4**.

Graph 1.4



I. Describe the effect of increasing the speed of moving air on the rate of water loss. [2]

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II. **Sketch a line on Graph 1.4** to suggest the result you would expect if the **humidity** of the air was increased. [1]

12



2. Image 2.1 shows chickens in two different farming systems.

Image 2.1



Free-range farm

Intensive farm

In 1950 chicken was an expensive food and most adults in the UK, on average, ate only 1100 g each year. By 2000 they ate 25 kg per year as intensive farming had made chicken much cheaper.

In intensive farming, large numbers of chickens are reared indoors. Environmental conditions and food supply are constantly controlled. The chickens grow faster than free-range chickens and use less energy as their movement is restricted.

The farmer can monitor the chickens more easily than on a free-range farm. Much less land is used and labour costs are lower but larger amounts of concentrated waste are produced.

- (a) (i) Use the information given to calculate the increase in mass in the annual consumption of chicken for a **family of four adults** in the UK between 1950 and 2000. [2]

Increase in mass = kg/family/year



During an investigation, the growth of chickens from different farm systems was compared. The results are shown in **Graph 2.2** and **Tables 2.3** and **2.4**.

Graph 2.2



Table 2.3

Organ mass in chickens at six weeks

Organ	Mass (g)	
	Free-range	Intensively farmed
Heart	6.5	4.8
Lungs	4.0	3.5
Liver	20.0	15.0

Table 2.4

Bone quality in chickens at six weeks

	Free-range	Intensively farmed
Bone density (g/cm ³)	1.29	0.79
Presence of broken bones (%)	2.5	37
Length of leg bones (mm)	73	118



Examiner
only

- (ii) Use **Graph 2.2** to calculate the difference in the body muscle mass at 5 weeks between intensively farmed chicken and a free-range chicken. [1]

Difference = g

- (b) (i) Use the information on page 7 to suggest **one** advantage to farmers of farming chickens intensively. [1]

.....

.....

- (ii) State **one** feature of intensive farming which is an environmental disadvantage. [1]

.....

- (c) Some groups of people have ethical objections to intensive farming because of animal welfare concerns.

Using the information in **Tables 2.3** and **2.4**, state **three** features of intensive farming which support this point of view. [3]

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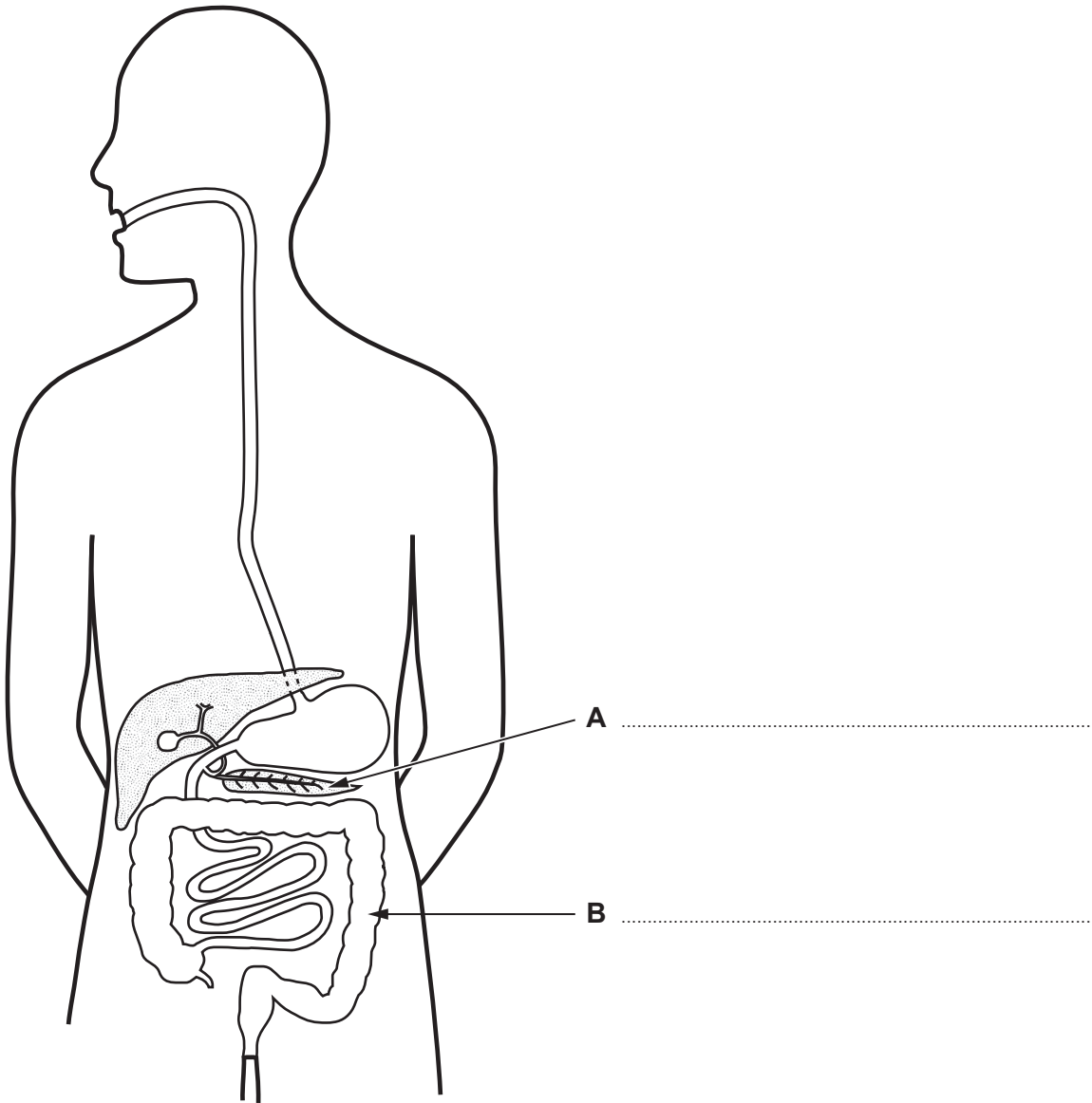
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3. **Image 3.1** shows the human digestive system.

Image 3.1



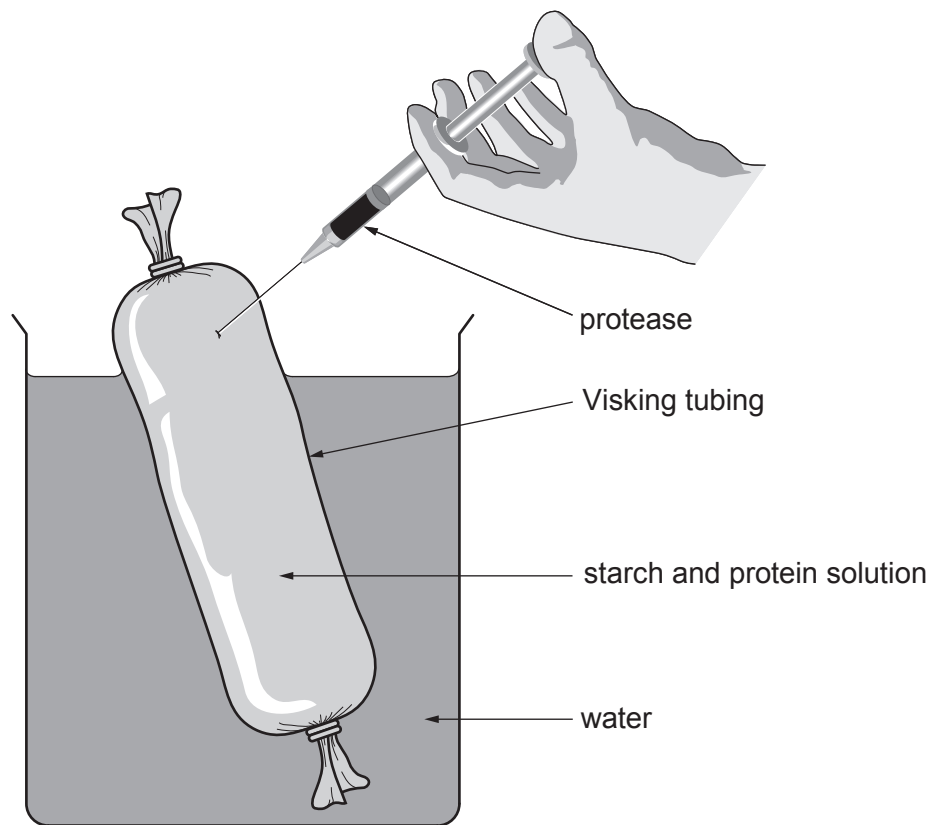
(a) Label parts **A** and **B** on **Image 3.1**.

[2]



- (b) A length of Visking tubing was used to model a part of the digestive system. The Visking tubing was filled with a solution of starch and protein. The apparatus was kept at a temperature of 35°C. The apparatus is shown in **Image 3.2**.

Image 3.2



A sample of the water surrounding the Visking tubing was tested for protein, starch, amino acids and glucose at the start of the experiment. After 15 minutes, protease was injected into the Visking tubing. After 30 minutes, the water surrounding the Visking tubing was tested again.

- (i) Describe how you would use a chemical to test the water for the presence of **glucose**.

[2]

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Examiner
only

(ii) **Complete Table 3.3** below to show the expected results for this experiment. One column has been done for you.

✓ = present in the water surrounding the Visking tubing
✗ = absent in the water surrounding the Visking tubing

[2]

Table 3.3

Time /minutes	Starch	Protein	Amino acids	Glucose
0	✗			
30	✗			

(c) Suggest which part of the digestive system shown in **Image 3.1** is represented by the Visking tubing. Give a reason for your answer.

[2]

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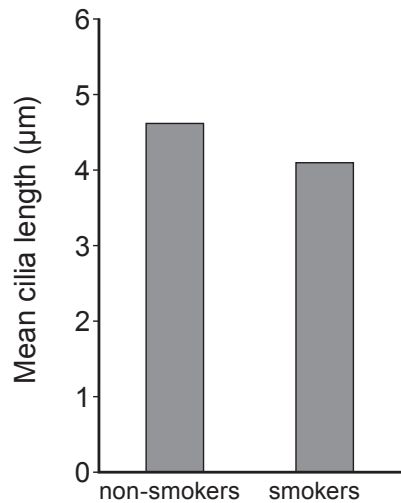
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4. A study of cilia length in the cells lining the bronchioles of 15 non-smokers and 15 smokers was carried out. The results are shown in **Graphs 4.1** and **4.2**.

Graph 4.1 mean cilia length in non-smokers and smokers



Graph 4.2 variation in cilia length in non-smokers and smokers

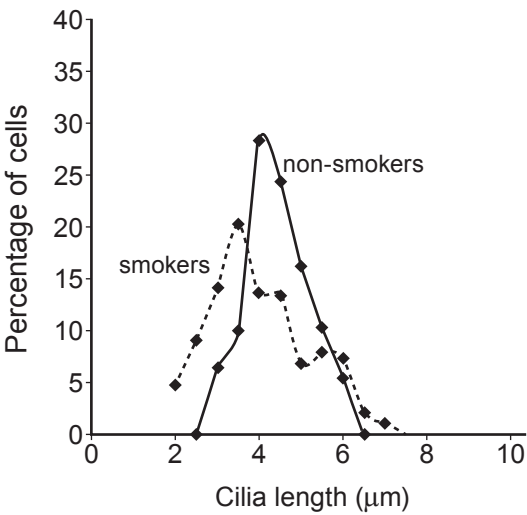
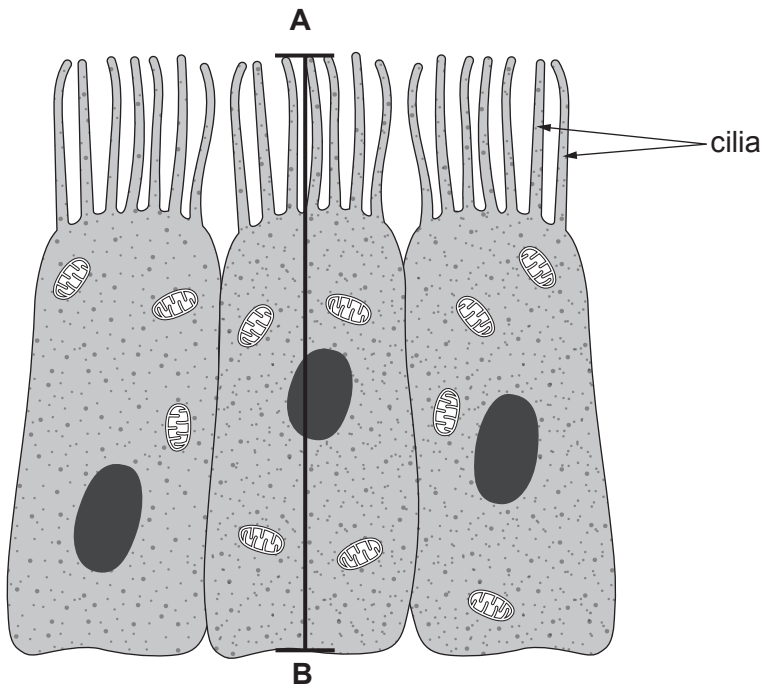


Image 4.3 shows three of the cells that line the bronchiole.

Image 4.3 – Cells lining the bronchioles.



Examiner
only

- (a) Describe the function of cilia in the human respiratory system and the effect of smoking on them. [2]

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- (b) (i) Measure the length of line **A–B** on **Image 4.3**. [1]

length **A–B** = mm

- (ii) The actual value of **A–B** in **Image 4.3** is $5\mu\text{m}$. Calculate the magnification of the drawing. (1 mm = $1000\mu\text{m}$) [2]

magnification = $\times$

- (c) (i) Using the information from **Graphs 4.1** and **4.2**, state **two** conclusions you can make about the effect of smoking on the length of cilia. [2]

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- (ii) Suggest **one** way in which you could improve the confidence in your conclusions. [1]

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5. According to a report by the British Heart Foundation in 2017, more than 20 million people in the UK were physically inactive. The report warned that inactivity increases the risk of heart disease and costs the NHS around £1.2 billion each year.
- A heart attack is a serious medical emergency in which the supply of blood to the heart is suddenly blocked, usually by a blood clot.

Image 5.1 shows the heart following a heart attack and **Image 5.2** shows a section through one of the heart's blood vessels.

Image 5.1

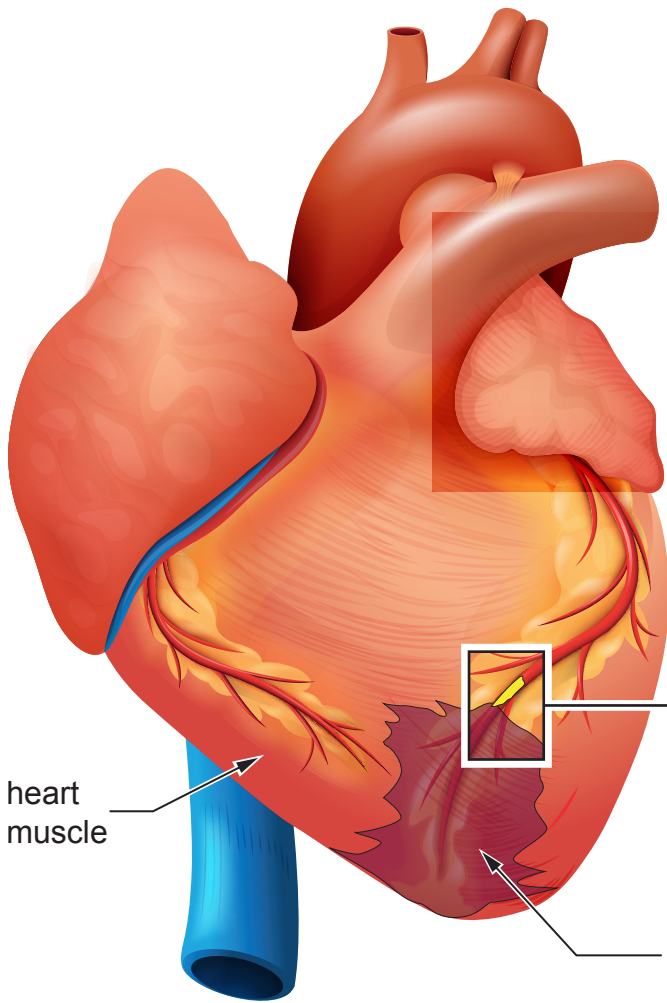
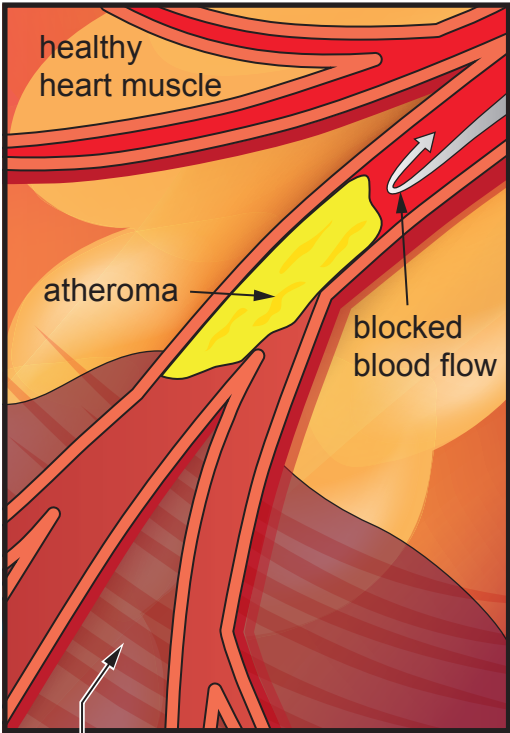


Image 5.2



- (a) (i) State the name of the blood vessel shown in **Image 5.2**. [1]
-
- (ii) State which part of blood is responsible for clot formation. [1]
-



Examiner
only

(b) (i) Explain how the atheroma has led to the death of heart muscle cells shown in **Images 5.1** and **5.2**. [3]

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(ii) Apart from lack of exercise, state **two** other factors which are linked to the increased risk of developing an atheroma. [2]

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(iii) State the name of a surgical procedure which can be used to remove blockages in the blood vessels surrounding the heart. [1]

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(iv) Suggest **one** way in which the Welsh Government could reduce the number of cases of heart disease each year. [1]

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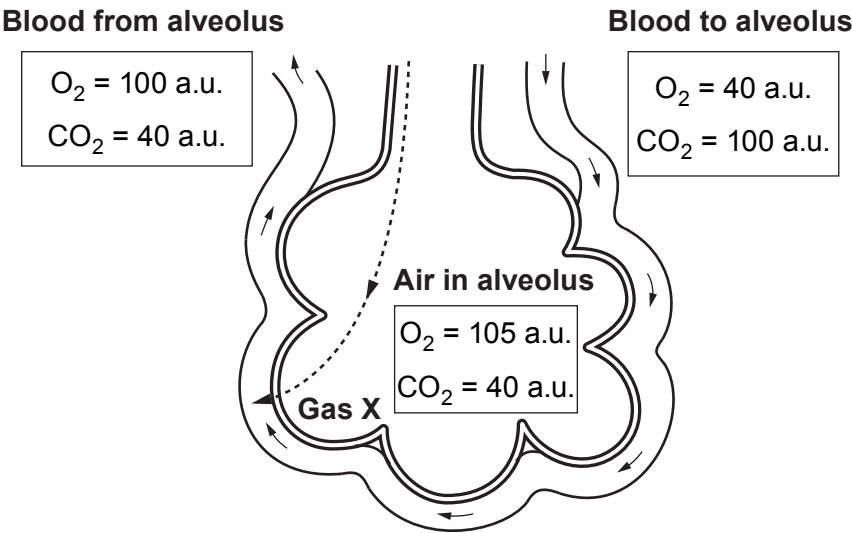
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6. **Image 6** shows gas exchange taking place between an alveolus and a blood capillary. The numbers shown represent the concentrations of oxygen (O_2) and carbon dioxide (CO_2) in arbitrary units (a.u.).

Image 6



(a) State **one** way in which the alveolus and capillary are adapted to their function. [2]

Adaptation of alveolus

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Adaptation of capillary

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(b) Using **Image 6**, name gas **X** and use the data to explain how this gas moves from the air in the alveolus into the blood. [3]

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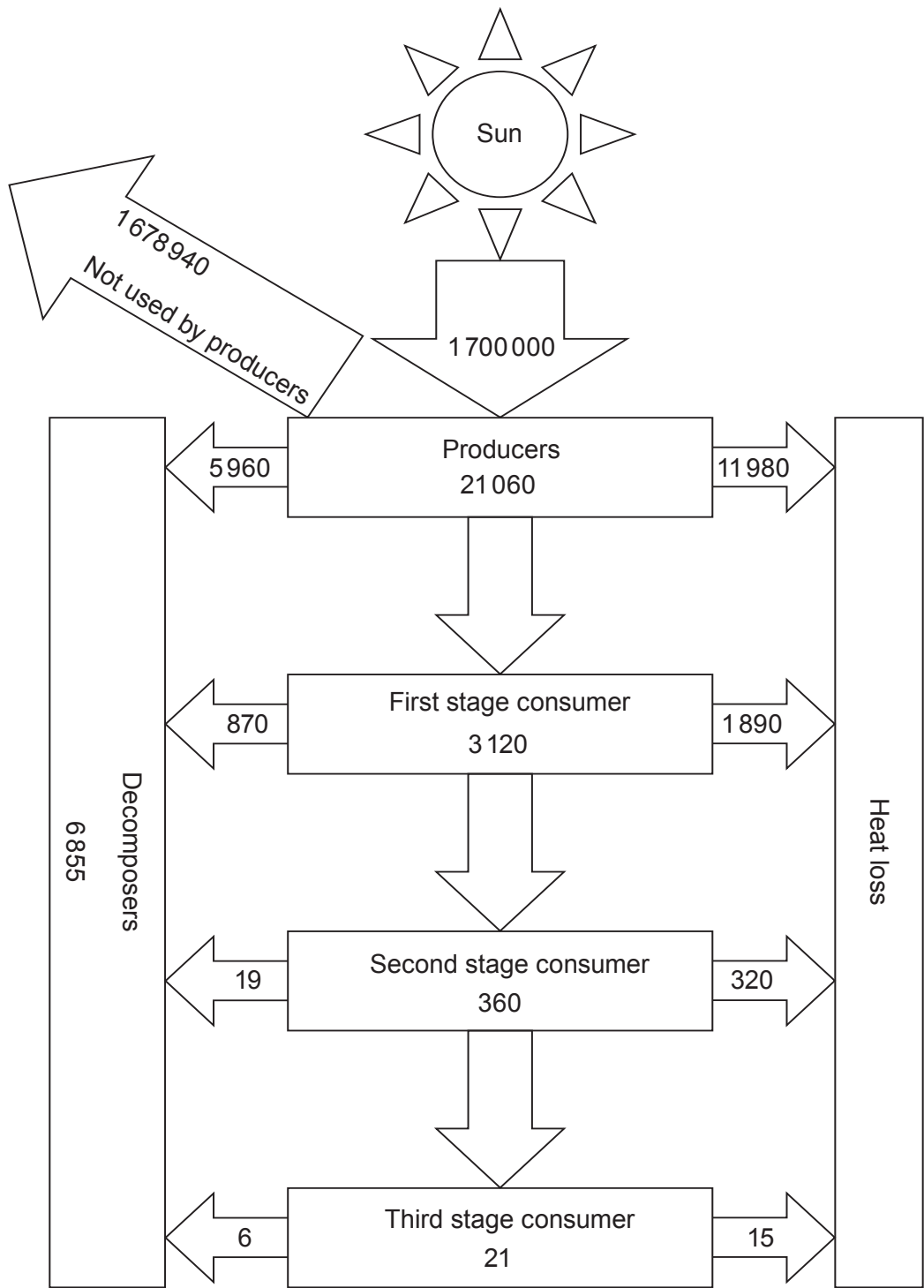
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7. **Image 7** shows the total amount of energy entering a grassland habitat in kilojoules per square metre during a year. The transfer of energy in a food chain is shown, as well as the energy that is lost from it as heat.

Image 7



Examiner
only

- (a)

(i)

State the name of the stage in **Image 7** which represents the herbivores.

[1]

.....

(ii)

State the name of **one** type of decomposer.

[1]

.....
- (b)

(i)

Use the data from **Image 7** to calculate the percentage of energy transferred from the first stage consumer to the second stage consumer. **Give your answer to three significant figures.**

[2]

Percentage of energy transferred =

- (ii)

Suggest why the food chain shown in **Image 7** would be unable to support a fourth stage consumer.

[1]
-
-



Examiner
only

(c) In the space below, **use a ruler** to draw a **labelled** pyramid of biomass to represent the food chain shown in **Image 7**. The pyramid does not need to be to scale. [1]

(d) State which process is responsible for the heat loss in **Image 7**. [1]

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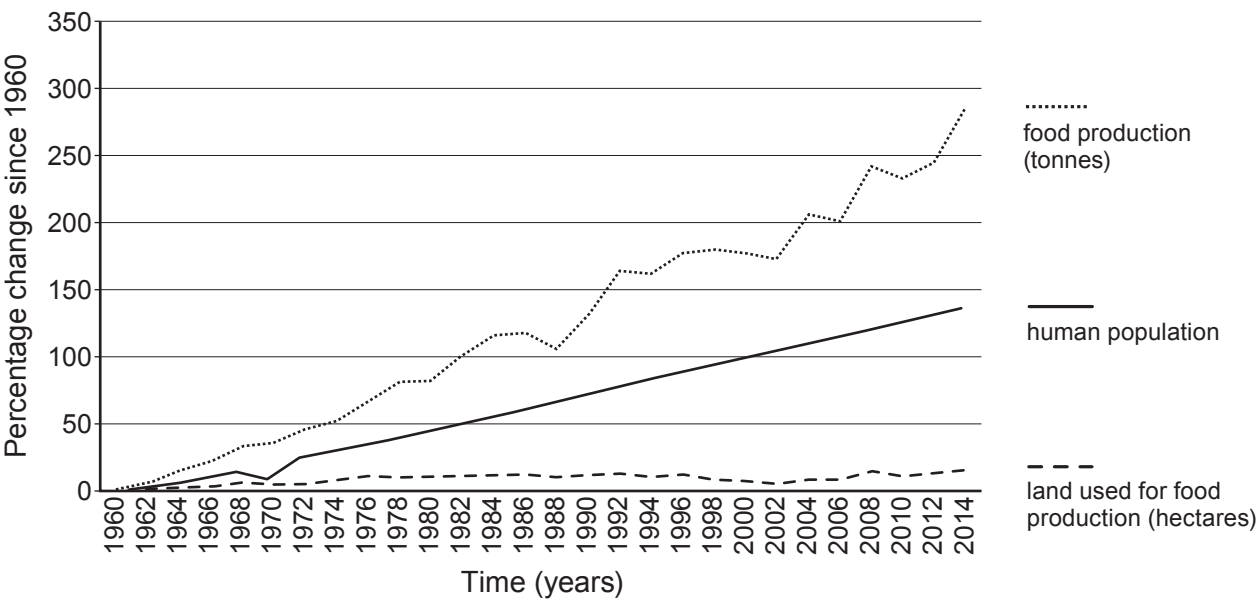
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Examiner
only

8. **Graph 8** shows the percentage change in global food production, human population and the land used for food production between 1960 and 2014.

Graph 8



(a) Use the information in **Graph 8** to state **one** conclusion about the efficiency of food production. Explain your answer.

[2]

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(b) Use your knowledge of the nitrogen cycle to explain how farmers can increase the nitrogen available to their crops and so increase their yield. Diagrams will not gain credit. [6 QER]

Examiner
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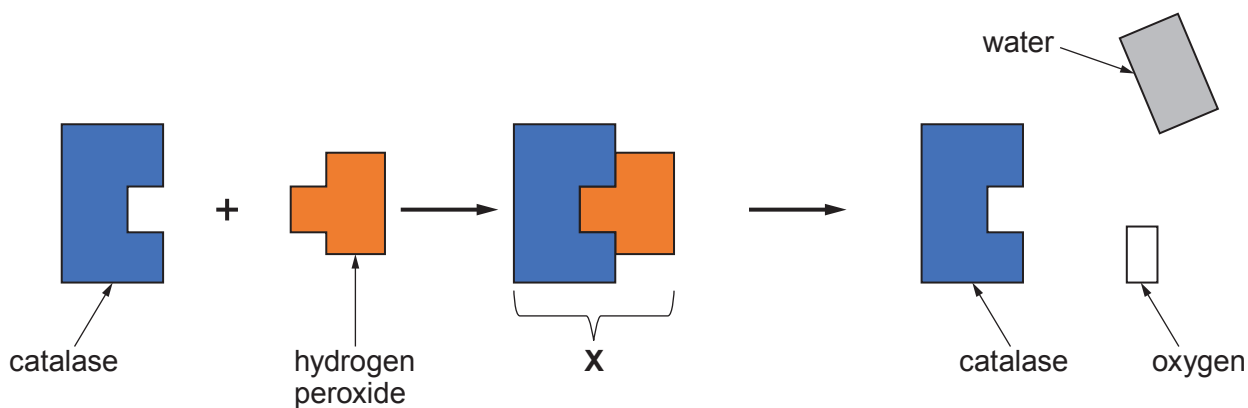
8



9. Catalase is an enzyme found in most cells. Catalase speeds up the breakdown of hydrogen peroxide to form water and oxygen.

Image 9.1 represents the breakdown of hydrogen peroxide by the action of catalase.

Image 9.1

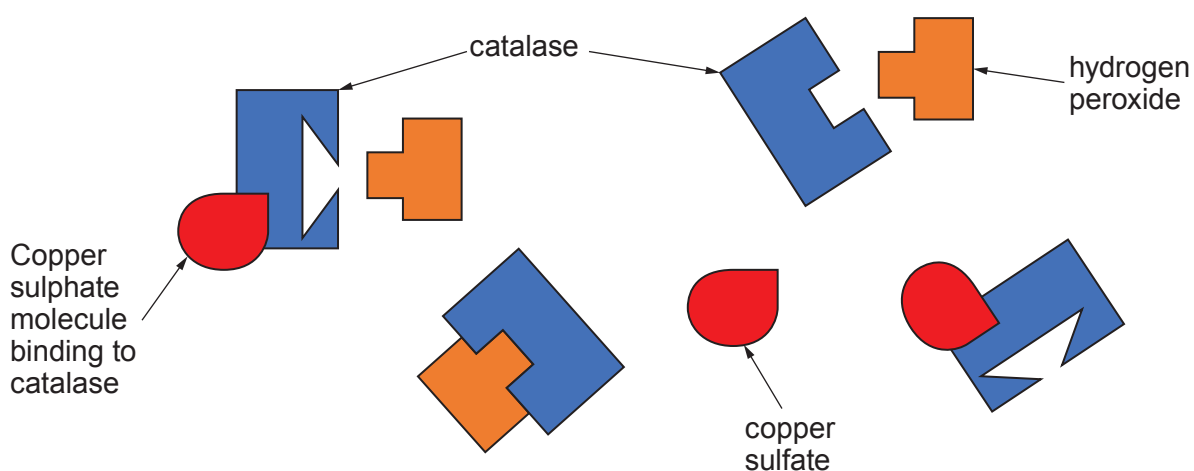


- (a) (i) State the name of the structure labelled **X** in **Image 9.1**. [1]

- (ii) State the name of the model of enzyme action shown in **Image 9.1**. [1]

- (b) **Image 9.2** represents the effect of copper sulfate on catalase. If a copper sulfate molecule binds to catalase it prevents it from working.

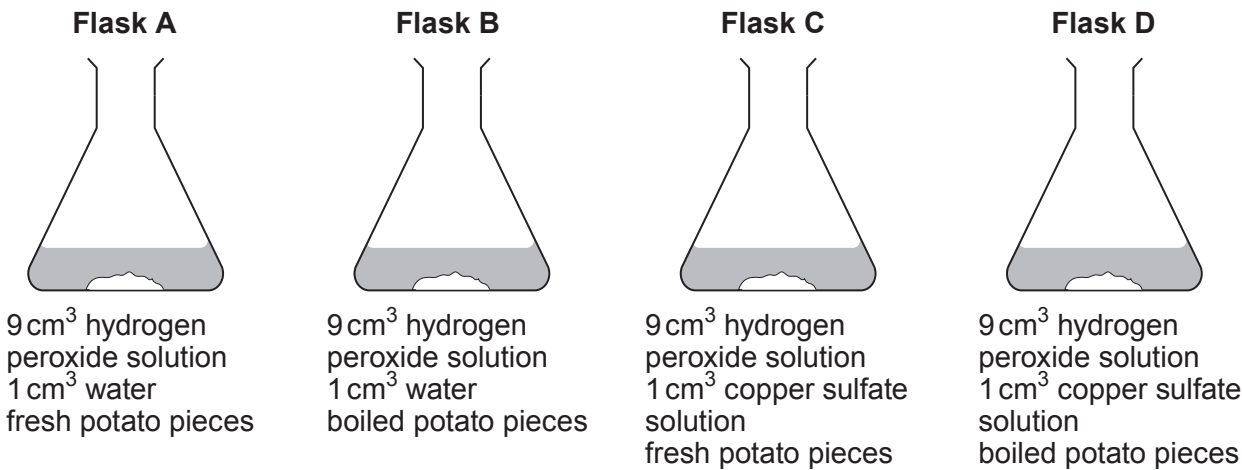
Image 9.2



Students investigated the effect of copper sulfate on catalase activity in potatoes.

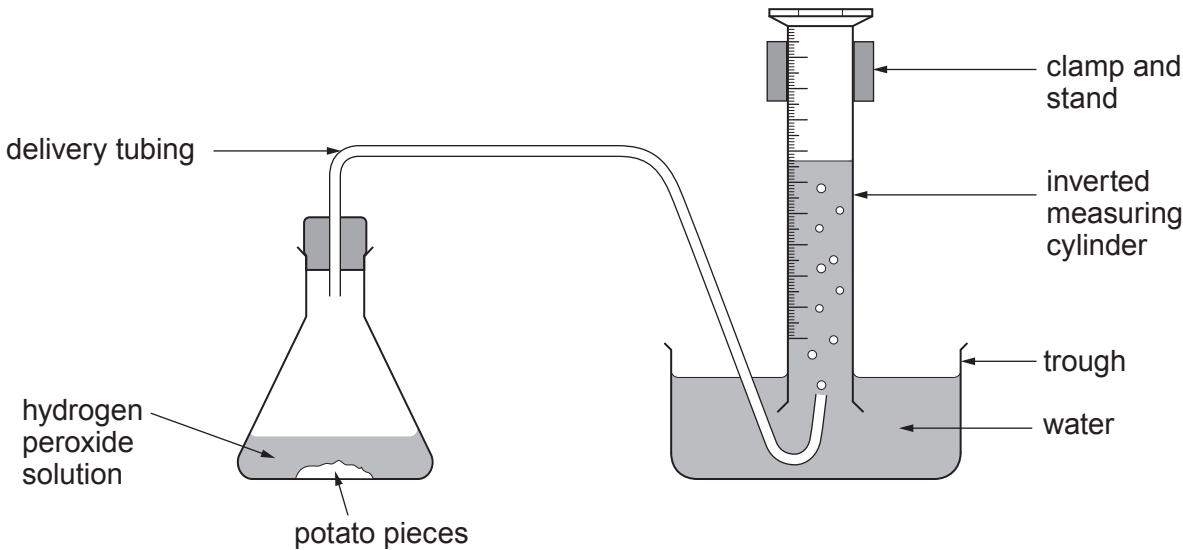
The students set up the following four flasks. as shown in **Image 9.3**.

Image 9.3



The students measured the volume of gas produced from each flask using the apparatus shown in **Image 9.4**.

Image 9.4



The results of the investigation are shown in **Table 9.5**.

Table 9.5

Time (minutes)	Volume of gas present in the cylinder (cm ³)			
	Flask A	Flask B	Flask C	Flask D
0	0	0	0	0
5	12	0	3	0
10	20	0	5	0
15	28	0	7	0
20	34	0	10	0
25	38	0	12	0
30	38	0	14	0

(i) Describe **and** explain the results for flask **A**. [4]

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(ii) The rate of gas production in flask **A** between 10 and 15 minutes is 1.6 cm³/min.
Calculate the rate of gas production between 10 and 15 minutes in flask **C**. [2]

Rate of gas production = cm³/minute



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(iii) Use the information in **Image 9.2** to explain why less gas is produced in flask **C** than in flask **A**. [3]

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(c) (i) State the purpose of flasks **B** and **D** in **Image 9.3**. [1]

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(ii) State **two** variables that should have been controlled in this investigation to make it a fair test. [2]

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(iii) Describe how the students could improve the accuracy of their investigation. [1]

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END OF PAPER

